

THÈSE PRÉSENTÉE
POUR OBTENIR LE GRADE DE
DOCTEUR
DE L'UNIVERSITÉ DE BORDEAUX

ÉCOLE DOCTORALE SCIENCES PHYSIQUES & DE L'INGÉNIEUR
AUTOMATIQUE, PRODUCTIQUE, SIGNAL ET IMAGE, INGÉNIERIE COGNITIVE

Par **Bastien HUBERT**

Cartographie et Navigation Autonome sur Champs Physiques par
Filtrage Particulaire à Processus Gaussiens

Autonomous Navigation and Mapping on Physical Fields via
Gaussian Process Particle Filtering

Sous la direction de : **Audrey GIREMUS**
la co-direction de : **Karim DAHIA**
et l'encadrement de : **Nicolas MERLINGE**

Thèse soutenue le 00 janvier 0000

Membres du jury :

Manon KOK	Professeure associée	TU Delft	Rapporteuse
Carl MILNER	Professeur associé	ENAC	Rapporteur
Vincent FREMONT	Professeur	Centrale Nantes	Examineur
Pierre MINVIELLE	Ingénieur de recherche	CEA	Examineur
Audrey GIREMUS	Professeure	Université de Bordeaux	Directrice
Karim DAHIA	Ingénieur de Recherche	ONERA	Co-directeur
Nicolas MERLINGE	Ingénieur de Recherche	ONERA	Encadrant

	ACKNOWLEDGEMENTS
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	ABSTRACTS
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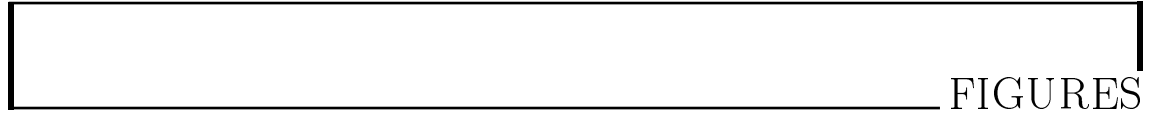
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ABBREVIATIONS

APF	Auxiliary Particle Filter
a.s.	almost surely
BLUE	Best Linear Unbiased Estimator
CDF	Cumulative Distribution Function
EKF	Extended Kalman Filter
GP	Gaussian Process
IEKF	Iterative Extended Kalman Filter
i.i.d.	independent and identically distributed
MC	Monte Carlo
MAP	Maximum <i>A Posteriori</i>
MISE	Mean Integrated Squared Error
ML	Maximum Likelihood
MMAE	Minimum Mean Absolute Error
MMSE	Minimum Mean Squared Error
PDF	Posterior Probability Density Function
PF	Particle Filter
RBPF	Rao-Blackwellised Particle Filter
RPF	Regularised Particle Filter
SIR	Sampling Importance Resampling
SIS	Sequential Importance Sampling
SPD	Symmetric Positive Definite
SPSD	Symmetric/Self-adjoint Positive Semi-Definite
TODO: finir la liste avec le chapitre 3	

SYMBOLS

$\triangleq$	Equality by definition
$\propto$	Proportionality
$\sim$	"is distributed according to"
$\overset{i.i.d.}{\sim}$	Independently and identically distributed according to
$\xrightarrow{a.s.}$	Almost sure convergence
$\xrightarrow{\mathcal{L}}$	Convergence in law
$\succeq 0$	Symmetric positive semi-definite
$\succ 0$	Symmetric positive definite
$\langle \cdot \cdot \rangle, \langle \cdot \cdot \rangle_{\mathcal{E}}$	Inner product, subscript: of the space $\mathcal{E}$
$\langle \cdot, \cdot \rangle$	Measure/function duality pairing, $\langle \mu, \varphi \rangle = \int \varphi d\mu$
$\ \cdot \ , \ \cdot \ _2, \ \cdot \ _F$	Generic, Euclidean and Frobenius norms
$\cdot^{\top}$	Transpose of a matrix
$\cdot^{-1}$	Inverse of a matrix
$*$	Convolution operator
$\circ$	Function composition operator
$\mathbb{1}$	Indicator function
$[\![\cdot, \cdot]\!]$	Discrete interval
$\cdot_{m n}$	"at time m conditioned on the observations from 1 to n "
α_{reg}	Bandwidth modulation factor
δ_x	Dirac measure at x
$\lambda_k^{(i)}$	Auxiliary weights
$\lambda_{\min}, \lambda_i$	Smallest and i -th eigenvalues of a matrix
μ	Target probability measure, of density p
$\mu_k^{(i)}$	Auxiliary particle
ν	Proposal (importance) measure, of density q
π_k	Projection of a trajectory onto its last component
ρ	Distance on the estimator space $\mathcal{F}$
φ	Mapping of the state
$\hat{\psi}_{\text{MMSE}}, \hat{\psi}_{\text{MMAE}}, \hat{\psi}_{\text{MAP}}, \hat{\psi}_{\text{ML}}$	State estimators
$\text{argmin}, \text{argmax}$	Argument of the minimum, of the maximum
$b_k^{(e)}, b_k^{(o)}$	Evolution and observation bias vectors
C_{μ}, C_{ν}	Normalising constants of μ and ν
cdf	Cumulative distribution function
d_x, d_z, d_{φ}	Dimensions of the state, observation, and mapping

d_w, d_n	Dimensions of the evolution and observation noises
$\mathbb{E}[\cdot], \mathbb{E}_p[\cdot]$	Expectation, subscript: under the law p
$\exp$	Exponential function
$f_k, f_k^l, f_k^n, \bar{f}_k^n$	Evolution functions
$F_k, F_k^l, F_k^n, \bar{F}_k^n$	Evolution matrices
$\mathcal{F}$	Space of admissible estimators
$\mathcal{GP}(m, k)$	Gaussian process
$h, h_{\text{Epa}}, h_{\text{Gauss}}$	Kernel bandwidths
h_k	Observation function
H_k	Observation matrix
I, I_n	Identity matrix, subscript: of size n
J_f	Jacobian matrix of a function f
k	Discrete time step
$k(\cdot, \cdot)$	Covariance kernel of a Gaussian process
$K, K_h, K_{\text{Epa}}, K_{\text{Gauss}}$	Regularisation kernels
K_k	Kalman gain
$\mathcal{L}^0, \mathcal{L}^p$	Sets of measurable functions and p -integrable random variables
m	Mean function of a Gaussian process
$\mathcal{M}_{m,n}(\mathbb{R})$	Set of real $m \times n$ matrices
n_k	Observation noise
N	Number of particles or samples
$\hat{N}_{\text{eff}}, N_{\text{eff}}$	Estimated and real effective sample size
N_{th}	Resampling threshold
$\mathbb{N}, \mathbb{N}^*$	Sets of natural and strictly positive natural integers
$\mathcal{N}(m, P)$	Gaussian distribution
$o(\cdot)$	Landau symbol for asymptotic negligibility
$\mathcal{O}(\cdot)$	Landau symbol for asymptotic boundedness
p	Target probability density
p_0	Probability density of the initial state
$p_k^{(e)}, p_k^{(o)}$	Probability densities of the evolution and observation noises
$p_{A B}(a b)$	Conditional density of A given B
$P_0, P_{k k-1}, P_{k k}$	State covariance matrices
$\mathbb{P}$	Probability of an event
q	Proposal (importance) density
$Q_k, Q_k^n, Q_k^l, Q_k^{nl}, \bar{Q}_k^l$	Evolution noise covariances
R_k	Observation noise covariance
$\mathbb{R}, \mathbb{R}_+, \mathbb{R}_+^*$	Sets of real, positive real, and strictly positive real numbers
$S^N(\mu), S^N(\mu, \nu), \tilde{S}^N, S_{K_h}^N$	Monte Carlo measures
S_k	Innovation covariance
supp	Support of a measure or function
tr	Trace of a matrix
$\mathcal{U}$	Input set of a stochastic process
$\mathcal{U}([a, b])$	Uniform distribution
$\mathbb{V}[\cdot], \mathbb{V}_\mu(\cdot)$	Variance, subscript: under the measure μ
$w_k, w_k^l, w_k^n, \bar{w}_k^l$	Evolution noises
$w_k^{(i)}, \tilde{w}_k^{(i)}$	Normalised and unnormalised particle weights
$x_k, x_k^n, x_k^l, \mathbf{x}_{0:k}$	System states and trajectory
$x_k^{(i)}$	Particle

$x_{k k-1}, x_{k k}, \hat{x}_k$		Predicted, corrected and estimated state
y_k		Innovation vector
$z_k, \mathbf{z}_{1:k}$		Observation vector and sequence
TODO: finir la liste avec le chapitre 3		

CHAPTER 1 PLAN GLOBAL

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1.1 Introduction

- contexte : navigation autonome, pas de GNSS mais des champs physiques
- motivation : limites des méthodes classiques (Kalman, carte embarquée connue)
- objectif : reconstruction de la fonction d'observation par GP, intégration dans le cadre de la navigation bayésienne
- principales contributions
- annonce du plan

Ne pas oublier la progression des problèmes:

1. drift of inertial navigation
2. GNSS data blurred or unavailable
3. high initial uncertainty
4. non-injective, non-linear, multimodal environments
5. undersampled, poorly-known environment model
6. weight degeneracy, particle impoverishment
7. non-stationarity of the field renders distant samples locally irrelevant
8. high computational cost of matrix inversion and multiplications
9. no preexisting data about the environment

1.2 Estimation bayésienne

- principes généraux et rappels de théorie de la mesure (cf cours de Nadia)
- filtre optimal
- filtres de Kalman et limites
- Kalman d'ensemble ?

1.3 Filtrage particulaire

- idem pour les maths
- filtrage particulaire simple et limites
- filtres particulaire avancés : régularisé, auxiliaire, marginalisé
- filtre de Kalman sur RKHS (ailleurs ?)

1.4 Processus gaussiens

- rappels de maths sur les processus stochastiques (s'inspirer Rasmussen)
- krigeage simple, ordinaire
- méthodes de réduction de rang
- lien avec les RKHS

1.5 Krigeage sous-échantillonné

- erreurs théoriques
- heuristiques de sous-échantillonnage (fenêtrage, clusterisation, sous-échantillonnage aléatoire)
- parler de l'intérêt pour la navigation ? permet de reconstruire la fonction d'observation ?
- résultats numériques sans navigation

1.6 CAK-PF

- fenêtrage, clustering par Mean-Shift
- apprentissage des hyperparamètres
- étude d'ablation
- précision en fonction du nombre de données disponibles
- résultats numériques (RMSE, temps de calcul, comparaisons avec la BCR)

1.7 SLAM

- GP stationnaires ou stationnaires par morceaux
- multi capteurs en mailles impossibles car dimensions trop proches
- RBPF/méthode de Manon
- RKHS/méthode de Karim

1.8 Ouverture sur la fusion de champs ?

- à voir mais information de Fisher théorique
- tests sur les 2 cas précédents

1.9 Conclusion et perspectives

- résumé des contributions : GP + FP
- adaptation du modèle en fonction de l'état (carte dégradée)
- reconstruction à partir d'un dictionnaire de fonctions de base (sans carte)
- limites des méthodes
- perspectives futures (modèles, méthodologie, ...)

Part I

Mathematical Foundations

CHAPTER 2

BAYESIAN ESTIMATION

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2.1 Introduction

The goal of this chapter is to formally introduce the tools necessary to solve the general filtering problem. More precisely, given a dynamical system whose state evolves according to a known stochastic model and is partially observed through noisy measurements, we seek to sequentially estimate the system state at each time step in an optimal sense. We first define what optimality means in a rigorous setting, then derive exact and approximate solutions to this problem. We also discuss the limitations of such methods.

Formally, for $d_x, d_z, d_\varphi \in \mathbb{N}^*$, we consider a hidden Markov chain $(x_k)_{k \geq 0}$ taking values in $\mathbb{R}^{d_x}$ and a series of observations $(z_k)_{k \geq 1}$ taking values in $\mathbb{R}^{d_z}$, each associated with the unknown state x_k . The quantity of interest is a mapping of the state under a known function $\varphi : \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_\varphi}$, which allows us to target any derived quantity of the state (*e.g.* position, velocity, or a subset of coordinates). When solving a navigation problem, the state and observation processes are linked by the state-space model [1]:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \quad (2.1)$$

where the processes $w_k \in \mathbb{R}^{d_w}$ and $n_k \in \mathbb{R}^{d_n}$ denote the evolution and observation noises respectively, and are assumed to be white noises (*i.e.* temporally independent, centered noises of finite variance), mutually independent and independent of x_0 .

The known data of this problem are:

$$\begin{cases} p_0 & \text{the probability density of the initial state } x_0 \\ f_k & \text{the evolution function} \\ p_k^{(e)} & \text{the probability density of the evolution noise } w_k \\ z_k & \text{the current noisy observation} \\ h_k & \text{the observation function} \\ p_k^{(o)} & \text{the probability density of the observation noise } n_k \end{cases} \quad (2.2)$$

Remark 2.1 (Control input). *In many navigation applications, a control input u_k is also available, and the evolution function takes the form $f_k(x_{k-1}, u_k, w_k)$. For the sake of clarity, we omit u_k from the general formulation as it can always be absorbed into f_k or treated as a known deterministic input.*

Depending on how observations are used relative to the time index of the state to estimate, three problems arise [2]:

- **Filtering:** estimate $\varphi(x_k)$ given $\mathbf{z}_{1:k} \triangleq (z_1, \dots, z_k)$. It is the most common case in real-time navigation.
- **Smoothing:** estimate $\varphi(x_l)$ for $l < k$ given $\mathbf{z}_{1:k}$. It corresponds to refining past estimates using future observations.
- **Prediction:** estimate $\varphi(x_l)$ for $l > k$ given $\mathbf{z}_{1:k}$. This amounts to anticipating future states from current observations.

This thesis focuses on the filtering problem. The rest of this chapter is organised as follows:

- Section 2.2 rigorously defines what it means to optimally estimate $\varphi(x_k)$ from the observations $\mathbf{z}_{1:k}$, and shows that all classical optimality criteria reduce to computing the posterior probability density function (PDF) $p(x_k|\mathbf{z}_{1:k})$.
- Section 2.3 derives analytical filtering algorithms to propagate this density over time, and discusses their limitations.
- Section 2.4 then presents asymptotically optimal numerical methods based on Monte Carlo (MC) sampling that alleviate such flaws.
- Finally, section 2.5 presents advanced particle filtering methods that improve upon standard MC techniques.

2.2 State Estimation

The first issue to tackle is to define precisely the meaning of "best estimator" of $\varphi(x_k)$ given $\mathbf{z}_{1:k}$ [3]. Formally, given a functional metric space $(\mathcal{F}, \rho)$, it amounts to finding a measurable function $\psi \in \mathcal{F}$ that minimises the expected loss:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\rho(\psi(\mathbf{z}_{1:k}), \varphi(x_k))) \quad (2.3)$$

Depending on the considered space and distance, different optimal estimators arise. We present three classical ones below. For the sake of simplicity, we write Z for a generic observation vector and X for the underlying state, as opposed to the time-indexed processes x_k and z_k . At a fixed time step k , (2.3) thus becomes:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\rho(\psi(Z), \varphi(X))) \quad (2.4)$$

For the remainder of this document, we let, for $d, d', p \in \mathbb{N}^*$:

$$\begin{aligned} \mathcal{L}^0(\mathbb{R}^{d'}, \mathbb{R}^d) &\triangleq \left\{ f : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d \mid f \text{ measurable} \right\} \\ \mathcal{L}^p(\mathbb{R}^d) &\triangleq \left\{ A \text{ an } \mathbb{R}^d\text{-valued random variable} \mid \mathbb{E}[\|A\|^p] < \infty \right\} \end{aligned} \quad (2.5)$$

where almost surely equal random variables are identified.

2.2.1 Posterior Expectation Estimator

Let:

$$\mathcal{F} \triangleq \left\{ \psi \in \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi}) \mid \mathbb{E}[\|\psi(Z)\|^2] < \infty \right\} \quad (2.6)$$

To simplify notations, we identify each estimator $\psi \in \mathcal{F}$ with the random variable $\psi(Z)$ it produces. Under this identification, $\mathcal{F}$ is a closed subspace of $\mathcal{L}^2(\mathbb{R}^{d_\varphi})$, and can be equipped with the inner product:

$$(A, B) \in \mathcal{F}^2 \mapsto \langle A|B \rangle \triangleq \mathbb{E}[A(Z)^\top B(Z)] \quad (2.7)$$

where $\mathbb{E}[X]$ denotes the expectation of the random variable X .

Let $\|\cdot\|$ be the norm associated with $\langle \cdot | \cdot \rangle$. When $\varphi(X) \in \mathcal{L}^2(\mathbb{R}^{d_\varphi})$, the minimisation problem (2.4) is equivalent to:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\mathbb{E}[\|\psi(Z) - \varphi(X)\|^2]) \quad (2.8)$$

An estimator solving (2.8) is called a minimum mean squared error (MMSE) estimator.

Proposition 2.1 (MMSE estimator). *The unique solution to (2.8) is the posterior expectation:*

$$\hat{\psi}_{\text{MMSE}}(z) \triangleq \mathbb{E}[\varphi(X)|Z = z] \quad (2.9)$$

Proof. Let $\hat{\psi}(z) \triangleq \mathbb{E}[\varphi(X)|Z = z]$ and $\psi \in \mathcal{F}$. It comes that:

$$\begin{aligned} \rho(\psi(Z), \varphi(X))^2 &= \langle \psi(Z) - \varphi(X) | \psi(Z) - \varphi(X) \rangle \\ &= \langle \psi(Z) - \hat{\psi}(Z) + \hat{\psi}(Z) - \varphi(X) | \psi(Z) - \hat{\psi}(Z) + \hat{\psi}(Z) - \varphi(X) \rangle \\ &= \langle \psi(Z) - \hat{\psi}(Z) | \psi(Z) - \hat{\psi}(Z) \rangle + \langle \hat{\psi}(Z) - \varphi(X) | \hat{\psi}(Z) - \varphi(X) \rangle \\ &\quad + 2\langle \psi(Z) - \hat{\psi}(Z) | \hat{\psi}(Z) - \varphi(X) \rangle \end{aligned} \quad (2.10)$$

The cross term vanishes by the tower property of conditional expectation:

$$\begin{aligned} \langle \psi(Z) - \hat{\psi}(Z) | \hat{\psi}(Z) - \varphi(X) \rangle &= \mathbb{E} \left[(\psi(Z) - \hat{\psi}(Z))^\top (\hat{\psi}(Z) - \varphi(X)) \right] \\ &= \iint (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \varphi(x)) p(dx, dz) \\ &= \iint (\psi(z) - \hat{\psi}(z))^\top (\hat{\psi}(z) - \varphi(x)) p(dx|z) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \left(\int (\hat{\psi}(z) - \varphi(x)) p(dx|z) \right) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \left(\hat{\psi}(z) - \mathbb{E}[\varphi(X)|Z = z] \right) p(dz) \\ &= \int (\psi(z) - \hat{\psi}(z))^\top \times 0 p(dz) \\ &= 0 \end{aligned} \quad (2.11)$$

by definition of $\hat{\psi}(z)$.

Therefore:

$$\begin{aligned} \rho(\psi(Z), \varphi(X))^2 &= \rho(\psi(Z), \hat{\psi}(Z))^2 + \rho(\hat{\psi}(Z), \varphi(X))^2 \\ &\geq \rho(\hat{\psi}(Z), \varphi(X))^2 \end{aligned} \quad (2.12)$$

with equality if and only if $\|\psi(Z) - \hat{\psi}(Z)\|^2 = 0$, i.e. $\psi = \hat{\psi}$ almost surely (a.s.). $\square$

Remark 2.2 (Geometric interpretation). *The MMSE estimator is the orthogonal projection of $\varphi(X)$ onto $\mathcal{F}$, viewed as a closed subspace of $\mathcal{L}^2(\mathbb{R}^{d_\varphi})$, i.e. the set of all square-integrable random variables that are measurable with respect to Z . The disappearance of the cross term in the proof above is precisely the orthogonality condition characterising this projection.*

Remark 2.3 (MMSE assumptions). *Square-integrability of $\varphi(X)$ is only required for the minimisation (2.8) to be well-posed, not the estimator itself. The posterior expectation $\mathbb{E}[\varphi(X)|Z = z]$ remains well-defined for any $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$, but it only minimises the mean squared error inside of $\mathcal{L}^2(\mathbb{R}^{d_\varphi})$.*

2.2.2 Posterior Median Estimator

Because the MMSE estimator minimises a quadratic loss, it is highly sensitive to extreme values and may perform poorly for heavy-tailed distributions. To mitigate this sensitivity to outliers, one can use an absolute loss instead. In dimension $d_\varphi = 1$, let:

$$\mathcal{F} \triangleq \{\psi \in \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}) \mid \mathbb{E}[|\psi(Z)|] < \infty\} \quad (2.13)$$

As previously, $\mathcal{F}$ is identified as a subspace of $\mathcal{L}^1(\mathbb{R})$, and is equipped with the following non-Euclidean distance:

$$(A, B) \in \mathcal{F}^2 \mapsto \rho(A, B) \triangleq \mathbb{E}[|A(Z) - B(Z)|] \quad (2.14)$$

The minimisation problem (2.4) thus becomes:

$$\operatorname{argmin}_{\psi \in \mathcal{F}} (\mathbb{E}[|\psi(Z) - \varphi(X)|]) \quad (2.15)$$

While the terminology is less standard than MMSE, an estimator solving (2.15) is often called a minimum mean absolute error (MMAE) estimator.

Proposition 2.2 (MMAE estimator, simplified assumptions). *Assume that, for every z , the conditional distribution of $\varphi(X)$ given $Z = z$ is absolutely continuous with a strictly positive density. Then, the unique solution to (2.15) is the posterior median of $\varphi(X)$ given $Z = z$, i.e. the value $\hat{\psi}_{\text{MMAE}}(z) \in \mathbb{R}$ that satisfies:*

$$\mathbb{P}(\varphi(X) \geq \hat{\psi}_{\text{MMAE}}(z) | Z = z) = \mathbb{P}(\varphi(X) < \hat{\psi}_{\text{MMAE}}(z) | Z = z) = \frac{1}{2} \quad (2.16)$$

Proof. By the law of total expectation, for any $\psi \in \mathcal{F}$, we have:

$$\mathbb{E}[|\psi(Z) - \varphi(X)|] = \mathbb{E} \left[\int |\psi(z) - \varphi(x)| p(dx | Z = z) \right] \quad (2.17)$$

Since the outer expectation is over non-negative quantities, this is minimised by minimising the inner integral pointwise for each z , i.e. by solving:

$$\operatorname{argmin}_{u \in \mathbb{R}} \int |\varphi(x) - u| p(dx | Z = z) \quad (2.18)$$

As $u \mapsto |\varphi(x) - u|$ is convex, so is the integral, hence any stationary point is a global minimum. Since the conditional law is absolutely continuous, the integral is differentiable in u . By differentiating this expression with respect to $u \in \mathbb{R}$, it follows that:

$$\begin{aligned} & \frac{d}{du} \left(\int |\varphi(x) - u| p(dx | Z = z) \right) = 0 \\ \iff & \frac{d}{du} \left(\int ((\varphi(x) - u) \mathbb{1}_{\varphi(x) \geq u}(x) - (\varphi(x) - u) \mathbb{1}_{\varphi(x) < u}(x)) p(dx | Z = z) \right) = 0 \\ \iff & \int (-\mathbb{1}_{\varphi(x) \geq u}(x) + \mathbb{1}_{\varphi(x) < u}(x)) p(dx | Z = z) = 0 \\ \iff & \mathbb{P}(\varphi(X) \geq u | Z = z) = \mathbb{P}(\varphi(X) < u | Z = z) \end{aligned} \quad (2.19)$$

Since the total probability is one, this is equivalent to $u = \hat{\psi}_{\text{MMAE}}(z)$ as defined in (2.16), with the strict positivity of the density ensuring uniqueness. $\square$

Remark 2.4 (Median assumptions). *The absolute continuity and positivity assumptions are only required for the clean statement above, not for the existence of a solution. Since the inner integral is convex, continuous and coercive in u , a minimum always exists for any integrable conditional distribution. In full generality, the set of solutions is the closed interval of medians:*

$$\left\{ u \in \mathbb{R} \mid \mathbb{P}(\varphi(X) \leq u \mid Z = z) \geq \frac{1}{2} \text{ and } \mathbb{P}(\varphi(X) \geq u \mid Z = z) \geq \frac{1}{2} \right\} \quad (2.20)$$

The absence of atoms then guarantees that this interval is characterised by the equality (2.16), while the strict positivity of the density ensures that it reduces to a single point.

Remark 2.5 (Generalised median). *The extension to higher dimensions $d_\varphi > 1$ is non-trivial. The $\mathcal{L}^1(\mathbb{R}^{d_\varphi})$ median, also called geometric median, is the most common generalisation. In the remainder of this thesis, we focus on the MMSE and MAP estimators (presented below), which extend naturally to any dimension.*

2.2.3 Maximum *a Posteriori* Estimator

As they produce a single point estimate that may lie in a low-probability region, both the MMSE and the MMAE estimators may not be suited for multimodal posteriors. In such cases, a natural alternative is to select a maximum of the posterior density.

Let $\mathcal{F} \triangleq \mathcal{L}^0(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi})$, with no integrability constraint. The maximum *a posteriori* (MAP) estimator is defined as the global mode of the posterior density of $\varphi(X)$:

$$\hat{\psi}_{\text{MAP}} \triangleq \operatorname{argmax}_{\psi \in \mathcal{F}} (\mathbb{E} [p_{\varphi(X)|Z}(\psi(Z)|Z)]) \quad (2.21)$$

As the objective is separable over z , the maximisation can be carried out pointwise, yielding for each z :

$$\hat{\psi}_{\text{MAP}}(z) = \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{\varphi(X)|Z}(u|z)) \quad (2.22)$$

Using Bayes' theorem, this is equivalent to:

$$\hat{\psi}_{\text{MAP}}(z) = \operatorname{argmax}_{u \in \mathbb{R}^{d_\varphi}} (p_{Z|\varphi(X)}(z|u) p_{\varphi(X)}(u)) \quad (2.23)$$

since the normalising constant $p_Z(z)$ is independent of u .

Although the MAP estimator avoids low-density regions, it still reduces the posterior to a single point. In particular, in multimodal situations, only one mode is retained, while the remaining ones are discarded.

Remark 2.6 (MAP optimisation problem). *The MAP estimator can be seen as the solution of an optimisation problem of the form (2.4), associated with the threshold loss:*

$$\forall \varepsilon > 0, (A, B) \in \mathcal{F}^2 \mapsto \rho_\varepsilon(A, B) \triangleq \mathbb{1}_{\|A(Z) - B(Z)\| > \varepsilon} \quad (2.24)$$

in the limit $\varepsilon \rightarrow 0^+$:

$$\hat{\psi}_{\text{MAP}} = \lim_{\varepsilon \rightarrow 0^+} \operatorname{argmin}_{\psi \in \mathcal{F}} (\rho_\varepsilon(\psi(Z), \varphi(X))) \quad (2.25)$$

Maximum Likelihood Estimator

In the special case where no prior distribution on $\varphi(X)$ is assumed, (2.23) reduces to the maximum likelihood (ML) estimator:

$$\hat{\psi}_{\text{ML}}(z) \triangleq \underset{u \in \mathbb{R}^{d_\varphi}}{\operatorname{argmax}} (p_{Z|\varphi(X)}(z|u)) \quad (2.26)$$

As it does not take any prior information on $\varphi(X)$ into account, this estimator is not a Bayesian estimator and may be poorly suited when the available data is scarce. While it coincides with the MAP estimator under a uniform prior, it will not be considered in the rest of this document.

2.3 Analytical Density Estimation

As established in section 2.2, all three estimators ultimately rely on the PDF $p(x_k|\mathbf{z}_{1:k})$, from which an estimate of $\varphi(x_k)$ can be derived. Point estimators provide a concise and directly interpretable representation of the state, which is often desirable for our applications. However, reducing the posterior density to a single point estimate may discard valuable information, particularly when it is heavy-tailed or multimodal.

The goal of the following section is thus to present algorithms that propagate the PDF itself at each discrete time step k in order to solve the inference problem presented by (2.1).

2.3.1 Optimal Bayesian Filtering Algorithm

In the Bayesian filtering framework, the PDF can be propagated recursively through two steps. However, each step relies on a specific conditional independence hypotheses as presented by Assumptions 2.1 and 2.2.

Assumption 2.1 (Markov property). *The state sequence $(x_k)_{k \geq 0}$ is a Markov chain. As such, it verifies:*

$$p(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) = p(x_k|x_{k-1}) \quad (2.27)$$

with $\mathbf{x}_{0:k} \triangleq (x_0, \dots, x_k)$.

Assumption 2.2 (Conditional observation independence). *The current observation z_k only depends on the current state x_k :*

$$p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) = p(z_k|x_k) \quad (2.28)$$

Both assumptions follow from the state-space model (2.1) and, in particular, from the whiteness of w_k and n_k and their independence from x_0 . Building on them, the optimal estimation algorithm computes the PDF in two steps: prediction and correction.

Prediction. The prior density at time k can be obtained by marginalising the previous PDF:

$$\begin{aligned} p(x_k|\mathbf{z}_{1:k-1}) &= \int p(x_k, x_{k-1}|\mathbf{z}_{1:k-1}) dx_{k-1} \\ &= \int p(x_k|x_{k-1}, \mathbf{z}_{1:k-1}) p(x_{k-1}|\mathbf{z}_{1:k-1}) dx_{k-1} \end{aligned} \quad (2.29)$$

Assumption 2.1 simplifies (2.29) into the Chapman-Kolmogorov equation [4]:

$$p(x_k | \mathbf{z}_{1:k-1}) = \int p(x_k | x_{k-1}) p(x_{k-1} | \mathbf{z}_{1:k-1}) dx_{k-1} \quad (2.30)$$

where $p(x_k | x_{k-1})$ is the evolution, or transition, density induced by the state equation $x_k = f_k(x_{k-1}, w_k)$ and the density $p_k^{(e)}$ of the evolution noise w_k .

Correction. Upon receiving the new observation z_k , Bayes' theorem gives:

$$p(x_k | \mathbf{z}_{1:k}) = \frac{p(z_k | x_k, \mathbf{z}_{1:k-1}) p(x_k | \mathbf{z}_{1:k-1})}{p(z_k | \mathbf{z}_{1:k-1})} \quad (2.31)$$

Applying Assumption 2.2 to (2.31) yields:

$$p(x_k | \mathbf{z}_{1:k}) = \frac{p(z_k | x_k) p(x_k | \mathbf{z}_{1:k-1})}{\int p(z_k | x_k) p(x_k | \mathbf{z}_{1:k-1}) dx_k} \quad (2.32)$$

where the likelihood $p(z_k | x_k)$ depends on the observation model $z_k = h_k(x_k, n_k)$ and the density $p_k^{(o)}$ of the observation noise n_k .

These two equations allow for a recursive algorithm to compute the PDF of the system at each time step. This algorithm, usually referred to as the optimal Bayesian filter [5], is given by Algorithm 2.1.

Algorithm 2.1 Optimal Bayesian Filter

Compute $p(x_k \mathbf{z}_{1:k-1})$ using (2.30)	▷ Prediction
Compute $p(x_k \mathbf{z}_{1:k})$ using (2.32)	▷ Correction

While theoretically optimal, this algorithm has no closed-form solution in general. The integrands in both (2.30) and (2.32) are analytically intractable for arbitrary f_k, h_k .

2.3.2 Kalman Filter

While the optimal estimation algorithm cannot be used in the form given by Algorithm 2.1, a closed-form solution can be obtained by introducing additional hypotheses on (2.1), known as the Kalman assumptions [6]. In this section, we assume that:

$$\left\{ \begin{array}{l} x_0 \sim \mathcal{N}(\bar{x}_0, P_0) \\ f_k : x_{k-1} \mapsto F_k x_{k-1} + b_k^{(e)} \\ w_k \sim \mathcal{N}(0, Q_k), \text{ independent of } x_0 \\ z_k \text{ is known} \\ h_k : x_k \mapsto H_k x_k + b_k^{(o)} \\ n_k \sim \mathcal{N}(0, R_k), \text{ independent of } x_0 \text{ and } w_k \end{array} \right. \quad (2.33)$$

where $F_k \in \mathcal{M}_{d_x, d_x}(\mathbb{R})$ and $H_k \in \mathcal{M}_{d_z, d_x}(\mathbb{R})$ are known matrices, and $b_k^{(e)} \in \mathbb{R}^{d_x}$ and $b_k^{(o)} \in \mathbb{R}^{d_z}$ are known bias vectors. $P_0 \succ 0$ is the initial state covariance, and $Q_k \succ 0$ and $R_k \succ 0$ are the evolution and observation noise covariance matrices. Here, $\succ 0$ denotes the symmetric positive definiteness (SPD) of a matrix [7], as introduced by Definition 2.1.

Definition 2.1 (Matrix symmetric positive definiteness). Let $n \in \mathbb{N}^*$ and $M \in \mathcal{M}_{n,n}(\mathbb{R})$. M is symmetric positive semi-definite (SPSD), denoted $M \succeq 0$, if it is:

$$\begin{cases} M^\top = M & \text{symmetric} \\ \forall a \in \mathbb{R}^n, a^\top M a \geq 0 & \text{positive semi-definite} \end{cases} \quad (2.34)$$

If the inequality is strict for every non-zero vector, then M is SPD.

In particular, the state-space model (2.1) becomes:

$$\begin{cases} x_k = F_k x_{k-1} + b_k^{(e)} + w_k \\ z_k = H_k x_k + b_k^{(o)} + n_k \end{cases} \quad (2.35)$$

Finally, $x_0, (w_k)_{k \geq 0}$ and $(n_k)_{k \geq 0}$ are assumed to be mutually independent. Under these assumptions, the true state x_k follows a Gaussian distribution at each time step as an affine transformation of Gaussian random variables.

Proposition 2.3 (Gaussian closure under affine assumptions). For $m, n \in \mathbb{N}$, let the index $m|n$ denote "at time m conditioned on the observations from 1 to n ". Under the hypotheses of (2.33), if $p(x_{k-1}|\mathbf{z}_{1:k-1}) = \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$, then:

$$\begin{cases} p(x_k|\mathbf{z}_{1:k-1}) = \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1}) \\ p(x_k|\mathbf{z}_{1:k}) = \mathcal{N}(\hat{x}_{k|k}, P_{k|k}) \end{cases} \quad (2.36)$$

where the predicted and corrected states $\hat{x}_{k|k-1}$ and $\hat{x}_{k|k}$, and covariances $P_{k|k-1} \succ 0$ and $P_{k|k} \succ 0$ are given by Algorithm 2.2.

Proof. Prediction. Let $\mathbb{V}[X]$ denote the variance of the random variable X :

$$\mathbb{V}[X] \triangleq \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top] \quad (2.37)$$

Since $x_k = F_k x_{k-1} + b_k^{(e)} + w_k$ is an affine transformation of two independent Gaussian random variables $x_{k-1}|\mathbf{z}_{1:k-1} \sim \mathcal{N}(\hat{x}_{k-1|k-1}, P_{k-1|k-1})$ and $w_k \sim \mathcal{N}(0, Q_k)$, the predicted state $x_k|\mathbf{z}_{1:k-1}$ is also Gaussian, with:

$$\begin{aligned} \hat{x}_{k|k-1} &\triangleq \mathbb{E}[x_k|\mathbf{z}_{1:k-1}] \\ &= F_k \mathbb{E}[x_{k-1}|\mathbf{z}_{1:k-1}] + b_k^{(e)} + \mathbb{E}[w_k|\mathbf{z}_{1:k-1}] \\ &= F_k \hat{x}_{k-1|k-1} + b_k^{(e)} \end{aligned} \quad (2.38)$$

and:

$$\begin{aligned} P_{k|k-1} &\triangleq \mathbb{V}[x_k|\mathbf{z}_{1:k-1}] \\ &= F_k \mathbb{V}[x_{k-1}|\mathbf{z}_{1:k-1}] F_k^\top + \mathbb{V}[w_k|\mathbf{z}_{1:k-1}] \\ &= F_k P_{k-1|k-1} F_k^\top + Q_k \end{aligned} \quad (2.39)$$

The predicted covariance is thus SPD as the sum of an SPD matrix and the SPD matrix Q_k .

Correction. The joint distribution of (x_k, z_k) given $\mathbf{z}_{1:k-1}$ is also Gaussian, since $z_k = H_k x_k + b_k^{(o)} + n_k$ is an affine transformation of the independent Gaussian variables $x_k|\mathbf{z}_{1:k-1} \sim \mathcal{N}(\hat{x}_{k|k-1}, P_{k|k-1})$ and $n_k \sim \mathcal{N}(0, R_k)$. It comes that:

$$\begin{bmatrix} x_k \\ z_k \end{bmatrix} \Big| \mathbf{z}_{1:k-1} \sim \mathcal{N} \left(\begin{bmatrix} \hat{x}_{k|k-1} \\ H_k \hat{x}_{k|k-1} + b_k^{(o)} \end{bmatrix}, \begin{bmatrix} P_{k|k-1} & P_{k|k-1} H_k^\top \\ H_k P_{k|k-1} & S_k \end{bmatrix} \right) \quad (2.40)$$

where:

$$S_k \triangleq H_k P_{k|k-1} H_k^\top + R_k \quad (2.41)$$

is called the innovation covariance, and is the covariance associated with the innovation vector:

$$y_k \triangleq z_k - \left(H_k \hat{x}_{k|k-1} + b_k^{(o)} \right) \quad (2.42)$$

Since $R_k \succ 0$, the independent pair (x_k, n_k) has the covariance:

$$\begin{bmatrix} P_{k|k-1} & 0 \\ 0 & R_k \end{bmatrix} \succ 0 \quad (2.43)$$

Its image under the invertible linear map $(x, n) \mapsto (x, H_k x + n)$ therefore has an SPD covariance as well. The Gaussian conditioning formula (see Proposition A.1) can thus be applied to give the posterior distribution $x_k | \mathbf{z}_{1:k}$ is Gaussian, with:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k \quad (2.44)$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (2.45)$$

where:

$$K_k \triangleq P_{k|k-1} H_k^\top S_k^{-1} \quad (2.46)$$

is the Kalman gain.

Finally, Proposition A.1 proves that the Schur complement of the SPD joint covariance:

$$P_{k|k-1} - (P_{k|k-1} H_k^\top) S_k^{-1} (H_k P_{k|k-1}) = P_{k|k-1} - K_k H_k P_{k|k-1} = P_{k|k} \quad (2.47)$$

is itself SPD, so $P_{k|k} \succ 0$. $\square$

The resulting algorithm, known as the Kalman filter [8], is depicted by Figure 2.1 and presented in Algorithm 2.2.

Algorithm 2.2 Kalman Filter

$\hat{x}_{k k-1} = F_k \hat{x}_{k-1 k-1} + b_k^{(e)}$	$\triangleright$ State prediction
$P_{k k-1} = F_k P_{k-1 k-1} F_k^\top + Q_k$	$\triangleright$ Covariance prediction
$y_k = z_k - \left(H_k \hat{x}_{k k-1} + b_k^{(o)} \right)$	$\triangleright$ Innovation
$S_k = H_k P_{k k-1} H_k^\top + R_k$	$\triangleright$ Innovation covariance
$K_k = P_{k k-1} H_k^\top S_k^{-1}$	$\triangleright$ Kalman gain
$\hat{x}_{k k} = \hat{x}_{k k-1} + K_k y_k$	$\triangleright$ State correction
$P_{k k} = (I - K_k H_k) P_{k k-1}$	$\triangleright$ Covariance correction

Remark 2.7 (Kalman state estimation). *Under the Kalman assumptions, the PDF is Gaussian, and so every Bayesian estimator presented in section 2.2 coincides with $\hat{x}_{k|k}$:*

$$\hat{\psi}_{\text{MMSE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\text{MMAE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k}) = \hat{x}_{k|k} \quad (2.48)$$

However, since the above equations use a predictive prior, the ML estimator does not generally correspond to the corrected state.

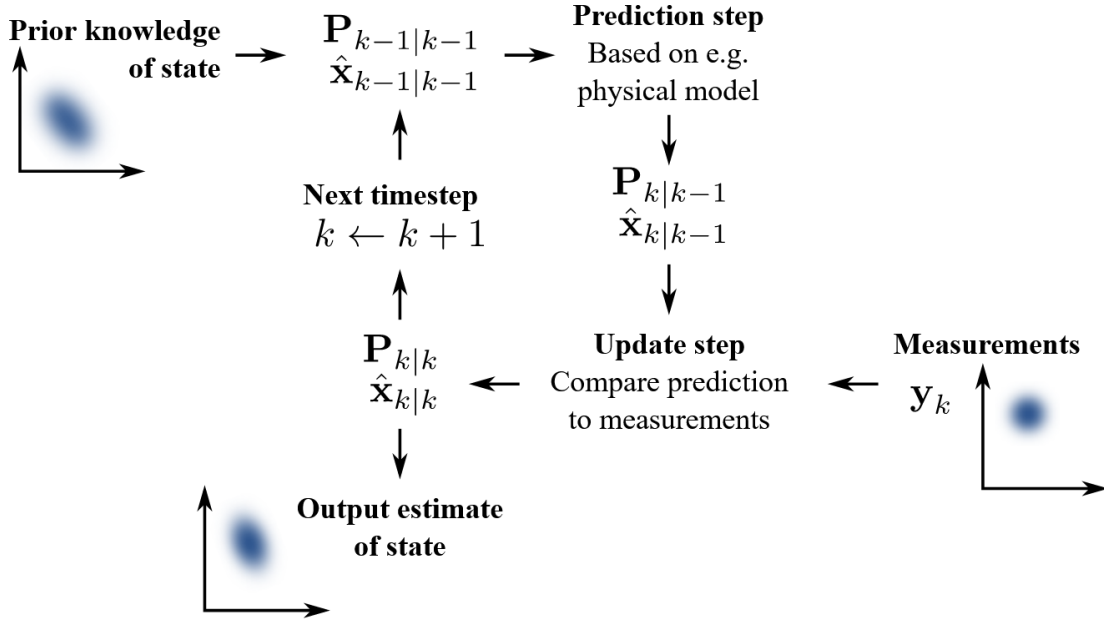


Figure 2.1: Schematic representation of the Kalman filter prediction/correction cycle. Each ellipsoid represents a confidence region associated with its corresponding Gaussian density.

Remark 2.8 (Precomputation of gains). *Under the assumptions of (2.33), the matrices $P_{k|k-1}$, K_k and $P_{k|k}$ depend only on the system matrices F_k , H_k , Q_k , R_k and the initial covariance P_0 , but not on the observations z_k . They can therefore be computed before any measurement is received.*

Joseph's Formula

In finite-precision arithmetic, the standard update $P_{k|k} = (I - K_k H_k) P_{k|k-1}$ may lose symmetry or positive definiteness due to rounding errors. Joseph's formula [9, 10] gives an alternative way to compute $P_{k|k}$ while ensuring that this property remains numerically satisfied:

$$\begin{aligned}
 P_{k|k} &= (I - K_k H_k) P_{k|k-1} \\
 &= (I - K_k H_k) P_{k|k-1} - K_k S_k K_k^\top + K_k S_k K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} - (P_{k|k-1} H_k^\top S_k^{-1}) S_k K_k^\top + K_k (H_k P_{k|k-1} H_k^\top + R_k) K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} - P_{k|k-1} H_k^\top K_k^\top + K_k H_k P_{k|k-1} H_k^\top K_k^\top + K_k R_k K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} - (I - K_k H_k) P_{k|k-1} H_k^\top K_k^\top + K_k R_k K_k^\top \\
 &= (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^\top + K_k R_k K_k^\top
 \end{aligned} \tag{2.49}$$

Under this more robust form, $P_{k|k}$ is SPSPD as the sum of two SPSPD matrices.

2.3.3 Extended Kalman Filter

The hypothesis that the evolution and observation functions are affine in x_k is often too restrictive in practice. The extended Kalman filter (EKF) [11] relaxes this

assumption by locally linearising f_k and h_k around the current state estimate via a first-order Taylor expansion.

Let $m, n \in \mathbb{N}^*$, let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at some $a \in \mathbb{R}^n$. f admits the following first-order expansion:

$$\forall x \in \mathbb{R}^n, f(x) = f(a) + J_f(a) (x - a) + o(\|x - a\|_2) \quad (2.50)$$

where $J_f(a) \in \mathcal{M}_{m,n}(\mathbb{R})$ is the Jacobian matrix of f evaluated at a , and defined as:

$$J_f \triangleq \frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f^{(1)}}{\partial x^{(1)}} & \cdots & \cdots & \frac{\partial f^{(1)}}{\partial x^{(n)}} \\ \vdots & \ddots & \ddots & \vdots \\ \frac{\partial f^{(m)}}{\partial x^{(1)}} & \cdots & \cdots & \frac{\partial f^{(m)}}{\partial x^{(n)}} \end{bmatrix} \quad (2.51)$$

and $f^{(i)}$ and $x^{(j)}$ respectively denote the i -th and j -th component of f and x .

Assuming that f_k and h_k are differentiable along the state trajectory, we can apply the first-order expansion (2.50) to the evolution and observation model. Successive linearisations at each time step can then be used to apply the Kalman filter equations locally. The resulting algorithm is given by Algorithm 2.3.

Algorithm 2.3 Extended Kalman Filter

$\tilde{F}_k = J_{f_k}(\hat{x}_{k-1 k-1})$	▷ Evolution Jacobian
$\hat{x}_{k k-1} = f_k(\hat{x}_{k-1 k-1}) + b_k^{(e)}$	▷ State prediction
$P_{k k-1} = \tilde{F}_k P_{k-1 k-1} \tilde{F}_k^\top + Q_k$	▷ Covariance prediction
$\tilde{H}_k = J_{h_k}(\hat{x}_{k k-1})$	▷ Observation Jacobian
$y_k = z_k - (h_k(\hat{x}_{k k-1}) + b_k^{(o)})$	▷ Innovation
$S_k = \tilde{H}_k P_{k k-1} \tilde{H}_k^\top + R_k$	▷ Innovation covariance
$K_k = P_{k k-1} \tilde{H}_k^\top S_k^{-1}$	▷ Kalman gain
$\hat{x}_{k k} = \hat{x}_{k k-1} + K_k y_k$	▷ State correction
$P_{k k} = (I - K_k \tilde{H}_k) P_{k k-1}$	▷ Covariance correction

While less restrictive than the standard Kalman filter, the EKF is suboptimal for three reasons:

- First, f_k and h_k must be differentiable along the state trajectory, which excludes discontinuous or piecewise-defined models.
- Second, the linearisation in (2.50) neglects higher-order terms of the Taylor expansion, which introduces an approximation error that grows with the degree of nonlinearity of f_k and h_k . This error increases over time as it propagates through the equations of the Kalman filter due to the sequential formulation.
- Finally, it relies on Gaussian assumptions and thus fails to handle multimodal or heavy-tailed distributions, as does its affine counterpart.

2.3.4 Other Suboptimal Kalman Filters

Iterative EKF

The iterative EKF (IEKF) [12] improves upon the EKF correction step by iteratively re-linearising h_k around successive state estimates within the same measurement update, while keeping the prediction step unchanged. Starting from the initialisation $\hat{x}_{k|k}^{(0)} = \hat{x}_{k|k-1}$, each iteration produces a new $\hat{x}_{k|k}^{(i)}$ by repeating the correction step from the predicted state, with the Jacobian evaluated at the latest estimate. This is done until:

$$\left\| \hat{x}_{k|k}^{(i)} - \hat{x}_{k|k}^{(i-1)} \right\|_2 \leq \varepsilon \quad (2.52)$$

for some $\varepsilon > 0$, or a maximum number of iterations is reached. The corrected state and covariance matrix are then defined as the result of the last loop.

This procedure is equivalent to performing a Gauss-Newton descent [13] on the posterior density maximisation problem, and converges to the MAP estimator under Gaussian assumptions. It reduces the linearisation error of the correction step when h_k is strongly non-linear, at the cost of several Jacobian evaluations per time step.

Unscented Kalman Filter

TODO+teaser le kernel Kalman filter [14, 15]

2.4 Monte Carlo Methods

While Kalman-based extensions can alleviate some of the limitations of the original Kalman filter, they all enforce the PDF to be Gaussian at each time step. This assumption is often violated in practice, as navigation problems frequently involve highly non-linear and non-injective functions. In turn, these functions induce multimodal or heavy-tailed distributions, which cannot be adequately characterised by Gaussian hypotheses.

Monte Carlo (MC) methods [16, 17] offer an alternative approach that remains consistent under very mild assumptions. Their accuracy improves asymptotically with the number of samples, at the expense of an increased computational cost. This section introduces the main concepts for MC-based algorithms to solve the navigation problem (2.1).

2.4.1 Monte Carlo Integration

Even though MC methods originally stem from the computation of expectations, they approximate the entire distribution [18], from which every estimator of section 2.2 can be derived. Being itself an expectation, the MMSE estimator is simply the most immediate, and is where we begin.

Let X be a random variable taking values in $\mathbb{R}^{d_x}$ with probability density p , and let $\varphi \in \mathcal{L}^0(\mathbb{R}^{d_x}, \mathbb{R}^{d_\varphi})$ such that $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$. We wish to compute the expectation of $\varphi(X)$:

$$\mathbb{E}_p[\varphi(X)] = \int \varphi(x) p(x) dx \quad (2.53)$$

Let $\mu(dx) \triangleq p(x) dx$ be the probability measure with density p with respect to the Lebesgue measure. (2.53) can be rewritten as:

$$\langle \mu, \varphi \rangle \triangleq \int \varphi(x) \mu(dx) = \mathbb{E}_p[\varphi(X)] \quad (2.54)$$

Since μ is a continuous measure, evaluating (2.54) exactly is intractable in most practical situations. The MC method aims to approximate μ by an empirical measure induced by N independent samples, as presented by Definition 2.2 and Proposition 2.4.

Definition 2.2 (Empirical measure). *Let $N \in \mathbb{N}^*$ and $X^{(1)}, \dots, X^{(N)}$ be independent and identically distributed (i.i.d.) random variables drawn from μ . The empirical measure associated with $(X^{(i)})_{i \in [1, N]}$ is defined as:*

$$S^N(\mu) \triangleq \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}} \quad (2.55)$$

where δ_x denotes the Dirac measure at x .

Proposition 2.4 (Monte Carlo estimator). *The MC estimator:*

$$\langle S^N(\mu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \varphi(X^{(i)}) \quad (2.56)$$

is unbiased and consistent:

$$\mathbb{E}_p[\langle S^N(\mu), \varphi \rangle] = \mathbb{E}_p[\varphi(X)] = \langle \mu, \varphi \rangle \quad (2.57)$$

$$\langle S^N(\mu), \varphi \rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \quad (2.58)$$

If moreover $\varphi(X) \in \mathcal{L}^2(\mathbb{R}^{d_\varphi})$, its mean squared error is:

$$\mathbb{E}_p[|\langle S^N(\mu) - \mu, \varphi \rangle|^2] = \frac{1}{N} \text{tr } \mathbb{V}_\mu[\varphi] \quad (2.59)$$

where $\mathbb{V}_\mu[\varphi] \triangleq \langle \mu, (\varphi - \langle \mu, \varphi \rangle)(\varphi - \langle \mu, \varphi \rangle)^\top \rangle$ is the covariance matrix of φ under μ , and its rescaled error converges in distribution to a centred Gaussian:

$$\sqrt{N} \langle S^N(\mu) - \mu, \varphi \rangle \xrightarrow{\mathcal{L}} \mathcal{N}(0, \mathbb{V}_\mu[\varphi]) \quad (2.60)$$

Proof. By linearity of the expectation and the transfer theorem, it comes that:

$$\begin{aligned} \mathbb{E}_p[\langle S^N(\mu), \varphi \rangle] &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_p[\varphi(X^{(i)})] \\ &= \frac{1}{N} \left(\int \varphi(x) p(x) dx \right) \\ &= \frac{1}{N} \sum_{i=1}^N \langle \mu, \varphi \rangle \\ &= \langle \mu, \varphi \rangle \end{aligned} \quad (2.61)$$

Furthermore, by independence of the $X^{(i)}$:

$$\begin{aligned}
\mathbb{E}_p \left[\left| \langle S^N(\mu) - \mu, \varphi \rangle \right|^2 \right] &= \mathbb{E}_p \left[\left| \frac{1}{N} \sum_{i=1}^N (\varphi(X^{(i)}) - \langle \mu, \varphi \rangle) \right|^2 \right] \\
&= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}_p \left[\left| \varphi(X^{(i)}) - \langle \mu, \varphi \rangle \right|^2 \right] \\
&= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E} \left[\left| \varphi(X) - \langle \mu, \varphi \rangle \right|^2 \right] \\
&= \frac{1}{N} \text{tr } \mathbb{V}_\mu[\varphi]
\end{aligned} \tag{2.62}$$

Finally, the a.s. convergence and the convergence in distribution follow respectively from the strong law of large numbers and the central limit theorem, applied to the i.i.d. variables $\varphi(X^{(i)})$. $\square$

Informally, this means that:

$$S^N(\mu) = \frac{1}{N} \sum_{i=1}^N \delta_{X^{(i)}} \approx \mu \tag{2.63}$$

in the sense that, for any measurable function φ such that $\varphi(X)$ is integrable:

$$\langle S^N(\mu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \varphi(X^{(i)}) \approx \langle \mu, \varphi \rangle \tag{2.64}$$

2.4.2 Importance Sampling

The standard MC method requires sampling directly from μ , which can be complex in practice. Importance sampling [19] circumvents this issue by drawing from a chosen importance, or proposal, distribution ν with density q .

Assuming that $\text{supp}(q) \subseteq \text{supp}(p)$, and for $x \in \text{supp}(q)$, it follows from the identity $p(x) = \frac{p(x)}{q(x)} q(x)$ and the transfer theorem that:

$$\mathbb{E}_p[\varphi(X)] = \int \left(\varphi(x) \frac{p(x)}{q(x)} \right) q(x) dx = \mathbb{E}_q \left[\varphi(X) \frac{p(X)}{q(X)} \right] \tag{2.65}$$

Using measure theory notations, this is equivalent to writing:

$$\langle \mu, \varphi \rangle = \left\langle \nu, \frac{d\mu}{d\nu} \varphi \right\rangle \tag{2.66}$$

The density $\frac{d\mu}{d\nu}$ is known as the Radon-Nikodym derivative. This motivates the introduction of the weighted empirical measure, given by Definition 2.3.

Definition 2.3 (Weighted empirical measure). *Let $\nu(dx) = q(x) dx$ with $\text{supp}(\mu) \subseteq \text{supp}(\nu)$, and let $X^{(1)}, \dots, X^{(N)} \stackrel{i.i.d.}{\sim} \nu$. The associated weighted empirical measure is:*

$$S^N(\mu, \nu) \triangleq \frac{d\mu}{d\nu} S^N(\nu) = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \delta_{X^{(i)}} \tag{2.67}$$

Pairing this measure with a test function φ yields the weighted MC estimator of Proposition 2.5.

Proposition 2.5 (Weighted MC estimator). *The weighted MC estimator:*

$$\langle S^N(\mu, \nu), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \varphi(X^{(i)}) \quad (2.68)$$

is unbiased and consistent:

$$\mathbb{E}_p[\langle S^N(\mu, \nu), \varphi \rangle] = \mathbb{E}_p[\varphi(X)] \quad (2.69)$$

$$\langle S^N(\mu, \nu), \varphi \rangle \xrightarrow{a.s.} \mathbb{E}_p[\varphi(X)] \quad (2.70)$$

Furthermore, when $\mathbb{V}_\nu\left[\frac{d\mu}{d\nu}\varphi\right] < \infty$, its mean squared error and rescaled error satisfy:

$$\mathbb{E}_q\left[\left|\langle S^N(\mu, \nu) - \mu, \varphi \rangle\right|^2\right] = \frac{1}{N} \text{tr} \mathbb{V}_\nu\left[\frac{d\mu}{d\nu}\varphi\right] \quad (2.71)$$

$$\sqrt{N} \langle S^N(\mu, \nu) - \mu, \varphi \rangle \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \mathbb{V}_\nu\left[\frac{d\mu}{d\nu}\varphi\right]\right) \quad (2.72)$$

Proof. This is direct using the results of Proposition 2.4. $\square$

For $\varphi \equiv 1$, the total weight:

$$\langle S^N(\mu, \nu), 1 \rangle = \frac{1}{N} \sum_{i=1}^N \frac{p(X^{(i)})}{q(X^{(i)})} \quad (2.73)$$

converges a.s. to 1, but each finite sample is not necessarily normalised. Furthermore, p and q are often known only up to normalising constants $\tilde{p} = C_\mu p$ and $\tilde{q} = C_\nu q$. To alleviate these issues, Definition 2.4 introduces the self-normalised measure [20], whose estimator remains consistent despite these unknown constants, as proven by Proposition 2.6.

Definition 2.4 (Self-normalised measure). *Let $X^{(1)}, \dots, X^{(N)} \stackrel{i.i.d.}{\sim} \nu$, with $\tilde{\mu} = C_\mu \mu$ and $\tilde{\nu} = C_\nu \nu$. The self-normalised measure is:*

$$\tilde{S}^N(\mu, \nu) \triangleq \sum_{i=1}^N w^{(i)} \delta_{X^{(i)}} \quad (2.74)$$

$$w^{(i)} \triangleq \tilde{w}^{(i)} / \sum_{j=1}^N \tilde{w}^{(j)} \quad (2.75)$$

$$\tilde{w}^{(i)} \triangleq \frac{\tilde{p}(X^{(i)})}{\tilde{q}(X^{(i)})} = \frac{d\tilde{\mu}}{d\tilde{\nu}}(X^{(i)}) \quad (2.76)$$

Proposition 2.6 (Self-normalised estimator). *The self-normalised estimator:*

$$\langle \tilde{S}^N(\mu, \nu), \varphi \rangle = \sum_{i=1}^N w^{(i)} \varphi(X^{(i)}) = \frac{\langle S^N(\tilde{\mu}, \tilde{\nu}), \varphi \rangle}{\langle S^N(\tilde{\mu}, \tilde{\nu}), 1 \rangle} \quad (2.77)$$

is biased at finite N , but consistent as soon as $\varphi(X) \in \mathcal{L}^1(\mathbb{R}^{d_\varphi})$ and $\text{supp}(\mu) \subseteq \text{supp}(\nu)$:

$$\langle \tilde{S}^N(\mu, \nu), \varphi \rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \quad (2.78)$$

Proof. Although the bias of $\tilde{S}^N(\mu, \nu)$ admits no simple closed-form expression in general, it can be computed for $N = 1$. In this situation, the self-normalisation forces the single weight to 1, such that:

$$\langle \tilde{S}^1(\mu, \nu), \varphi \rangle = \varphi(X^{(1)}) \quad (2.79)$$

and thus:

$$\mathbb{E} \left[\langle \tilde{S}^1(\mu, \nu), \varphi \rangle \right] = \langle \nu, \varphi \rangle \neq \langle \mu, \varphi \rangle \quad (2.80)$$

in general.

Furthermore, applying the strong law of large numbers separately to both the numerator and denominator of the self-normalised estimator yields:

$$\langle S^N(\tilde{\mu}, \tilde{\nu}), \varphi \rangle = \frac{1}{N} \sum_{i=1}^N \tilde{w}^{(i)} \varphi(X^{(i)}) \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}} \varphi \right\rangle = \frac{C_\mu}{C_\nu} \langle \mu, \varphi \rangle \quad (2.81)$$

and:

$$\langle S^N(\tilde{\mu}, \tilde{\nu}), 1 \rangle = \frac{1}{N} \sum_{j=1}^N \tilde{w}^{(j)} \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}} \right\rangle = \frac{C_\mu}{C_\nu} \langle \mu, 1 \rangle = \frac{C_\mu}{C_\nu} \neq 0 \quad (2.82)$$

Combining both expressions finally gives the desired result. $\square$

2.4.3 Sequential Importance Sampling

In the filtering context, the target distribution is the joint posterior $p(\mathbf{x}_{0:k} | \mathbf{z}_{1:k})$ over the entire state trajectory. As new observations arrive, this distribution grows in dimension, making a static importance sampling approach impractical. Sequential importance sampling (SIS) [21, 22] addresses this by updating the weights recursively rather than re-approximating the full trajectory at every time step.

Assumption 2.3 (SIS hypotheses). *In addition to Assumptions 2.1 and 2.2, we assume that the proposal density of the previous state does not depend on the current observation:*

$$q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k}) = q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \quad (2.83)$$

This assumption enables the following factorisation:

$$\begin{aligned} q(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) &= q(x_k, \mathbf{x}_{0:k-1} | \mathbf{z}_{1:k}) \\ &= q(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k}) \\ &= q(x_k | \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \end{aligned} \quad (2.84)$$

While Assumption 2.3 is sufficient to derive the SIS algorithm, an additional assumption is often considered:

$$q(\mathbf{x}_{0:k} | \mathbf{z}_{1:k}) = q(x_k | x_{k-1}, z_k) q(\mathbf{x}_{0:k-1} | \mathbf{z}_{1:k-1}) \quad (2.85)$$

so that a new particle $x_k^{(i)}$ can be propagated from $x_{k-1}^{(i)}$ and z_k alone.

Using Assumptions 2.1 and 2.2 at time k , the PDF factorises as:

$$\begin{aligned}
p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) &= p(\mathbf{x}_{0:k}|z_k, \mathbf{z}_{1:k-1}) \\
&= \frac{p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\
&= \frac{p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(x_k, \mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\
&= \frac{p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) p(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\
&= \frac{p(z_k|x_k) p(x_k|x_{k-1}) p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\
&\propto p(z_k|x_k) p(x_k|x_{k-1}) p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})
\end{aligned} \tag{2.86}$$

Let $\mathbf{x}_{0:k}^{(i)}$ denote the sequence $(x_0^{(i)}, \dots, x_k^{(i)})$. The recursive weight update is obtained by combining (2.86) with Assumption 2.3:

$$\begin{aligned}
w_k^{(i)} &\propto \frac{\tilde{p}(\mathbf{x}_{0:k}^{(i)}|\mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k}^{(i)}|\mathbf{z}_{1:k})} \\
&\propto \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)}) \tilde{p}(\mathbf{x}_{0:k-1}^{(i)}|\mathbf{z}_{1:k-1})}{\tilde{q}(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}) \tilde{q}(\mathbf{x}_{0:k-1}^{(i)}|\mathbf{z}_{1:k-1})} \\
&\propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})}
\end{aligned} \tag{2.87}$$

with the further simplification under the hypothesis of (2.85):

$$w_k^{(i)} \propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)}|x_{k-1}^{(i)}, z_k)} \tag{2.88}$$

While the current formulation allows us to approximate the density of the full trajectory $p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ through the self-normalised measure of Definition 2.4, we often only want to estimate the current state x_k . This can be done naturally by letting π_k denote the projection of a $(k+1)$ -tuple onto its last element:

$$\pi_k : \begin{cases} (\mathbb{R}^{d_x})^{k+1} \rightarrow \mathbb{R}^{d_x} \\ (x_0, \dots, x_k) \mapsto x_k \end{cases} \tag{2.89}$$

and computing the self-normalised estimator for $\varphi = \pi_k$, which approximates the MMSE estimator of section 2.2:

$$\hat{x}_k \triangleq \left\langle \tilde{S}^N(\mu_k, \nu_k), \pi_k \right\rangle = \sum_{i=1}^N w_k^{(i)} x_k^{(i)} \approx \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k] \tag{2.90}$$

where $\mu_k(dx) \triangleq \tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) dx$ and $\nu_k(dx) \triangleq \tilde{q}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) dx$. The corresponding empirical covariance matrix is:

$$\hat{P}_k = \sum_{i=1}^N w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top \quad (2.91)$$

The other estimators of section 2.2 remain accessible, though they are not as immediate:

- For a scalar state, the MMAE estimator only requires the cumulative distribution function, which $\tilde{S}^N(\mu_k, \nu_k)$ does define, and is obtained by accumulating the weights of the sorted particles until the total exceeds $\frac{1}{2}$. Its extension to a vector state is however not straightforward.
- The MAP estimator, on the other hand, calls for a density, which the particle approximation does not possess. In particular, $\hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k})$ cannot be computed using the weights, since they are importance ratios rather than density values. The heaviest particle maximises the Radon-Nikodym derivative rather than the PDF itself, and maximising $\tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ over the particles only locates a mode among the sampled locations.

As for the optimal and Kalman filters, the SIS algorithm follows the prediction/correction loop. The prediction step corresponds to the sampling of the new particle set through the proposal density, and is often called particle propagation. Conversely, the correction step is the weight update of (2.87). The SIS algorithm is given by Algorithm 2.4.

Algorithm 2.4 Sequential Importance Sampling

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot x_{k-1}^{(i)}, z_k)$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k x_k^{(i)}) \tilde{p}(x_k^{(i)} x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} \mathbf{x}_{0:k-1}, \mathbf{z}_{1:k})}$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation

Remark 2.9 (Feynman-Kac distribution). *The recurrence (2.86) can be expanded:*

$$p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) \propto p(x_0) \prod_{j=1}^k p(x_j | x_{j-1}) \times \prod_{j=1}^k p(z_j | x_j) \quad (2.92)$$

to show that the PDF is a Feynman-Kac distribution [23, 24], with the transitions $p(x_j | x_{j-1})$ acting as the mutation and the likelihoods $p(z_j | x_j)$ as the selection function. Rather than being an artifice used to cancel the unknown constants, the self-normalisation introduced in Definition 2.4 is thus, more fundamentally, the normalisation of this distribution.

Numerical correction of the covariance matrix

For numerical reasons, the estimated covariance matrix $\hat{P}_k$ may not satisfy SPD properties in practice. While an analytic reformulation existed for the Kalman filter, the numerical nature of MC methods prevents a similar solution. Formally, we want to find the closest matrix $M_+^* \succ 0$ to $\hat{P}_k$ in the sense of the Frobenius norm:

$$A \in \mathcal{M}_{d_x, d_x}(\mathbb{R}) \mapsto \|A\|_F \triangleq \sqrt{\sum_{i=1}^{d_x} \sum_{j=1}^{d_x} A_{i,j}^2} \quad (2.93)$$

If $d_x = 1$, the positive real $m_+ \in \mathbb{R}_+$ closest to some real $m \in \mathbb{R}$ is:

$$m_+ = \frac{m + |m|}{2} \quad (2.94)$$

While this only applies to scalars, a similar operation can be performed in higher dimensions by using the SPSP term of the polar decomposition. The singular values decomposition is used to simplify the computation of the polar decomposition.

Formally, the correction algorithm [25] is as follows:

- Symmetrisation of $\hat{P}_k$: $M \triangleq \frac{\hat{P}_k + \hat{P}_k^\top}{2}$
- Singular values decomposition: $M = U D V^\top$
- Polar decomposition: $M = \Omega \Sigma$ with $\Omega \triangleq U V^\top$ and $\Sigma \triangleq V D V^\top$
- Computation of M_+ : $M_+ \triangleq \frac{M + \Sigma}{2}$

With this algorithm, M_+ is ensured to be SPSP. However, we wish to achieve strict positive definiteness. To solve this issue, we recall that a symmetric matrix is SPD if and only if all its eigenvalues are strictly positive, and we introduce a small correction $\varepsilon > 0$. Let $\lambda_{\min}$ denote the smallest eigenvalue of M_+ . Let:

$$\delta \triangleq \max(0, -\lambda_{\min}) + \varepsilon \quad (2.95)$$

where the first term is introduced to compensate any numerical errors introduced during the symmetrisation and SVD steps.

Since M_+ and δI commute, they are simultaneously diagonalisable, and the eigenvalues of:

$$M_+^* \triangleq M_+ + \delta I \quad (2.96)$$

are thus:

$$\lambda_i(M_+^*) = \lambda_i(M_+) + \delta \geq \lambda_{\min} + \delta > 0 \quad (2.97)$$

For small enough values of ε , M_+^* is the desired matrix.

2.4.4 Weight Degeneracy

A fundamental limitation of SIS is that the variance of the importance weights increases monotonically over time [26] as shown by Proposition 2.7.

Proposition 2.7 (Weight degeneracy). *Let $\tilde{w}_{0:k}$ denote the non-normalised weight associated with the entire trajectory:*

$$\tilde{w}_{0:k} \propto \frac{\tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}, \quad \mathbf{x}_{0:k} \sim q(\cdot|\mathbf{z}_{1:k}) \quad (2.98)$$

The variance:

$$\mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})}[\tilde{w}_{0:k}] \quad (2.99)$$

is non-decreasing in k , and equals zero if and only if $q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) = p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$.

Proof. TODO □

As k increases, the weight distribution typically becomes increasingly concentrated on a small number of particles, eventually leading to severe weight degeneracy. This implies that (2.74) is a poor approximation of the true PDF. While this is true in most cases, Proposition 2.8 exhibits a proposal density for which the degeneracy problem is reduced.

Proposition 2.8 (Optimal proposal density). *When the proposal density is the posterior transition:*

$$q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k|x_{k-1}, z_k) \quad (2.100)$$

the incremental weight variance:

$$\mathbb{V}_{q(\cdot|x_{k-1}^{(i)}, z_k)}[\tilde{w}_k^{(i)}] \quad (2.101)$$

is minimised, and the weight degeneracy is therefore delayed. However, it does not remove it in general.

Proof. Plugging (2.100) into (2.87) yields:

$$\begin{aligned} w_k^{(i)} &\propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})} \\ &\propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})}{\tilde{p}(x_k^{(i)}|x_{k-1}^{(i)}, z_k)} \\ &\propto w_{k-1}^{(i)} \frac{\tilde{p}(z_k|x_k^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})}{\tilde{p}(z_k|x_k^{(i)}, x_{k-1}^{(i)}) \tilde{p}(x_k^{(i)}|x_{k-1}^{(i)})} \\ &\quad \frac{\tilde{p}(z_k|x_{k-1}^{(i)})}{\tilde{p}(z_k|x_k^{(i)}, x_{k-1}^{(i)})} \\ &\propto w_{k-1}^{(i)} \tilde{p}(z_k|x_{k-1}^{(i)}) \frac{\tilde{p}(z_k|x_k^{(i)})}{\tilde{p}(z_k|x_k^{(i)}, x_{k-1}^{(i)})} \\ &\propto w_{k-1}^{(i)} \tilde{p}(z_k|x_{k-1}^{(i)}) \end{aligned} \quad (2.102)$$

By induction, it comes that $w_k^{(i)}$ does not depend on $x_k^{(i)}$. Thus:

$$\begin{aligned}
\mathbb{V}_{q(\cdot|x_{k-1}^{(i)}, z_k)} [\tilde{w}_k^{(i)}] &= \mathbb{E}_{q(\cdot|x_{k-1}^{(i)}, z_k)} [w_k^{(i)2}] - \mathbb{E}_{q(\cdot|x_{k-1}^{(i)}, z_k)} [w_k^{(i)}]^2 \\
&\propto \int w_k^{(i)2} q(x_k^{(i)}|x_{k-1}^{(i)}, z_k) dx_k^{(i)} - \left(\int w_k^{(i)} q(x_k^{(i)}|x_{k-1}^{(i)}, z_k) dx_k^{(i)} \right)^2 \\
&\propto w_k^{(i)2} \int q(x_k^{(i)}|x_{k-1}^{(i)}, z_k) dx_k^{(i)} - w_k^{(i)2} \left(\int q(x_k^{(i)}|x_{k-1}^{(i)}, z_k) dx_k^{(i)} \right)^2 \\
&\propto w_k^{(i)2} - w_k^{(i)2} \\
&= 0
\end{aligned} \tag{2.103}$$

□

While it reduces weight degeneracy, sampling from the posterior requires computing the marginal likelihood:

$$p(z_k|x_{k-1}^{(i)}) = \int p(z_k|x_k^{(i)}) p(x_k^{(i)}|x_{k-1}^{(i)}) dx_k^{(i)} \tag{2.104}$$

which is generally intractable.

A common alternative is to use the prior, or transition, density:

$$q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k|x_{k-1}) \tag{2.105}$$

which is easier to compute, but ignores the current observation during propagation.

The prior density also simplifies the weight update (2.87) to:

$$w_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k|x_k^{(i)}) \tag{2.106}$$

However, because the particles are propagated without regard to z_k , their incremental weight is simply the likelihood $\tilde{p}(z_k|x_k^{(i)})$. When this likelihood is peaked, most particles land where it is negligible. Only a few particles then carry most of the weight, which causes the degeneracy to accelerate. The prior is thus a less effective proposal in highly informative observation regimes.

2.4.5 Sampling Importance Resampling

Weight degeneracy is intrinsic to SIS and worsens with time unless the proposal distribution is chosen extremely close to the optimal proposal density. The sampling importance resampling (SIR) filter [27], also known as the bootstrap filter [28], mitigates this by introducing a resampling step after each weight update. Resampling replaces the weighted particle set $\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N$ by an unweighted set $\{x_k^{*(j)}, 1/N\}_{j=1}^N$ drawn according to a chosen resampling algorithm:

$$\{x_k^{*(j)}\}_{j=1}^N = \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right) \tag{2.107}$$

It also uses the transition density as proposal (2.105), which further reduces (2.106) to:

$$w_k^{(i)} \propto \tilde{p}(z_k | x_k^{(i)}) \quad (2.108)$$

since all weights are reset to $1/N$ after resampling. This makes the SIR applicable under very weak assumptions, only requiring the ability to sample from $p(\cdot | x_{k-1})$ and compute the likelihood $p(z_k | x_k)$. Its pseudo-code is given by Algorithm 2.5.

Algorithm 2.5 Sampling Importance Resampling

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim p(\cdot x_{k-1}^{(i)})$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = \tilde{p}(z_k x_k^{(i)})$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\{x_k^{(j)}, w_k^{(j)}, -\}_{j=1}^N \leftarrow \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right)$	▷ Resampling

Resampling Algorithms

Several resampling algorithms [29, 30] can be used in (2.107). We present three algorithms, which all select a new particle set from the particle approximation of the PDF:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} \delta_{x_k^{(i)}} \quad (2.109)$$

Drawing a particle from this discrete measure amounts to drawing its index $i \in \llbracket 1, N \rrbracket$ from the categorical distribution of the weights $(w_k^{(i)})_{i \in \llbracket 1, N \rrbracket}$ through its cumulative distribution function (CDF):

$$\begin{cases} \text{cdf}_i \triangleq \sum_{j=1}^i w_k^{(j)}, & i \in \llbracket 1, N \rrbracket \\ \text{cdf}_0 = 0 \end{cases} \quad (2.110)$$

As illustrated by Figure 2.2, a draw of $u_j \in [0, 1]$ uniquely determines the new particle $x_k^{*(j)}$ as the parent particle $x_k^{(i)}$ of index i satisfying:

$$\text{cdf}_{i-1} \leq u_j < \text{cdf}_i \quad (2.111)$$

The main difference between the three algorithms lies in how the N samples $\{u_j\}_{j \in \llbracket 1, N \rrbracket}$ are generated:

- **Multinomial resampling.** All the draws are independent:

$$u_j \stackrel{i.i.d.}{\sim} \mathcal{U}([0, 1]) \quad (2.112)$$

This is the most natural method, but has the highest resampling variance.

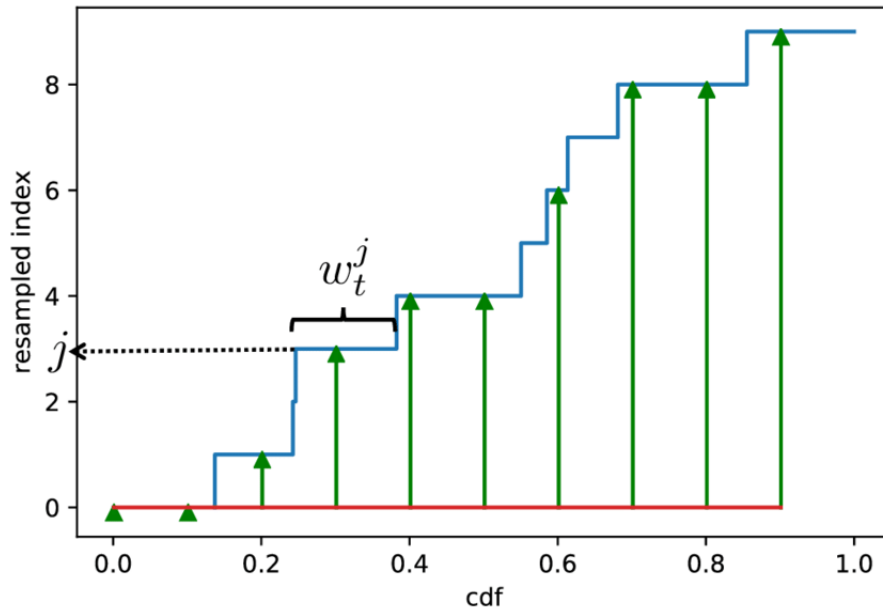


Figure 2.2: An example CDF of the particle weights. The width of each step is equal to the corresponding weight, so heavier particles are more likely to be selected.

- **Systematic or Kitagawa resampling [31].** A single draw generates all samples via:

$$\begin{cases} u_1 \sim \mathcal{U}([0, 1/N]) \\ u_j = u_1 + (j - 1)/N, j \in \llbracket 2, N \rrbracket \end{cases} \quad (2.113)$$

This significantly reduces the variance of the resampling step and is generally preferred in practice.

- **Deterministic resampling.** All samples are deterministic such that, for some $m \in [0, 1]$:

$$u_j = (j - m)/N, j \in \llbracket 1, N \rrbracket \quad (2.114)$$

This eliminates resampling variance entirely at the cost of a deterministic bias.

Some advanced filters, such as the auxiliary particle filter, presented in section 2.5, require keeping track of the parent index $i^{(j)}$ of each resampled particle. This value is thus recorded at the creation of each new particle. Algorithm 2.6 presents a linear implementation of the Kitagawa resampling algorithm, using the fact that both the CDF values and the Kitagawa samples are increasing to compute the parent indices sequentially.

Remark 2.10 (Loss of independence). *Because resampling duplicates particles, the resulting set is no longer independent, and the i.i.d. results of Propositions 2.4 and 2.5 no longer apply. However, the resampling is unbiased and the propagation step draws the particles independently conditionally on the past. The Feynman-Kac formalism proves [32, 33] that consistency and asymptotic Gaussianity still hold in this scenario, at the price of a variance larger than in the independent case and which accumulates over time.*

Algorithm 2.6 Kitagawa Resampling

```

cdf0 = 0                                ▷ Computation of the CDF
∀i ∈ [1, N], cdfi = cdfi-1 + wk(i)

u1 ~ U([0, 1/N])                        ▷ Kitagawa samples
∀j ∈ [2, N], uj = u1 + (j - 1)/N

i = 1
for j = 1 : N do
    while uj ≥ cdfi do                ▷ Computation of the parent index
        i ← i + 1
    end while
    xk*(j) = xk(i)                        ▷ Generation of the new particle
    wk*(j) = 1/N
    i(j) = i
end for

```

2.5 Advanced Particle Filters

Even though the SIR applies under very mild assumptions, it has several caveats. Indeed, while resampling at every step eliminates the weight degeneracy of the SIS, it wastes computational load, and erodes particle diversity. Furthermore, the number of particles required grows quickly with the state dimension, a phenomenon known as the curse of dimensionality [34].

The SIR admits several refinements [35], each targeting a specific issue. We first present the generic particle filter, which reduces the SIR's computational cost by only resampling when necessary. We then introduce the regularised and auxiliary particle filters to restore particle diversity and improve on the prior proposal. Finally, we derive the equations of the Rao-Blackwellised particle filter, which reduces the dimension of the explored space under linear-Gaussian substructure assumptions.

2.5.1 Generic Particle Filter

The SIR resamples at every time step, which eliminates weight degeneracy but comes with two drawbacks:

- It introduces unnecessary computational cost when weights are already well-distributed, and increases the variance of the estimated state, as mentioned by Remark 2.10.
- It also replicates the heaviest particles and discards the lightest, so the set progressively concentrates on a few distinct locations. This loss of diversity, known as particle impoverishment [36], is especially severe in low evolution noise or high-dimensional settings, where little diversity is re-introduced at the propagation step. It is illustrated by Figure 2.3.

To limit both issues, the generic particle filter (PF) [37] resamples only when the weight distribution has degraded significantly. This degradation is measured by the

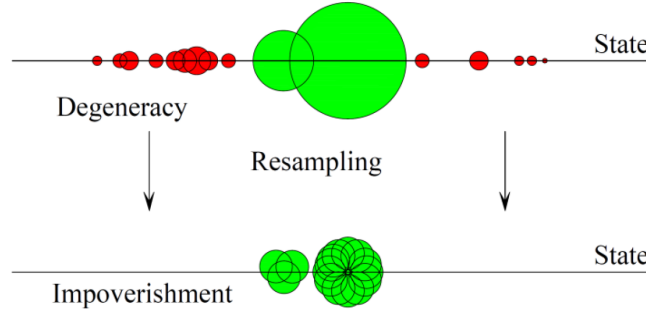


Figure 2.3: Particle impoverishment after resampling. Discarding non-resampled particles progressively shrinks the support of the particle approximation, degrading the filter’s ability to explore the state-space.

effective sample size:

$$N_{\text{eff}} \triangleq \frac{N}{1 + \mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})} \left[\frac{p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})} \right]} \quad (2.115)$$

Its denominator increases with the variance of the importance weights, whose growth over time corresponds to the weight degeneracy characterised by Proposition 2.7. When all the weights are uniform, the variance is zero, and thus $N_{\text{eff}} = N$. Conversely, the more unevenly they are distributed, the smaller N_{eff} , with $N_{\text{eff}} \rightarrow 1$ when the mass concentrates on a single particle.

Since the true importance weights are unknown, N_{eff} is estimated by replacing $\mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})}[\cdot]$ with the empirical spread of the normalised weights [38], which gives:

$$\hat{N}_{\text{eff}} \triangleq \frac{1}{\frac{1}{N} \sum_{i=1}^N \left(w_k^{(i)} \right)^2} \leq N \quad (2.116)$$

Like N_{eff} , this quantity ranges from N for uniform weights, down to 1 in the fully degenerate case. The PF then performs the resampling step only when this estimate falls below a threshold:

$$\hat{N}_{\text{eff}} < N_{\text{th}} \quad (2.117)$$

where N_{th} is a predetermined value, typically chosen proportional to N .

Unlike the SIR, which traditionally uses the transition density as its proposal distribution, the generic PF is not typically restricted to any particular choice of proposal distribution. It is presented by Algorithm 2.7.

2.5.2 Regularised Particle Filter

Since each resampling step still collapses diversity, the PF only mitigates impoverishment, but does not prevent it. Indeed, the self-normalised measure of Definition 2.4 approximates the PDF by a mixture of Diracs:

$$p(x_k|\mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} \delta_{x_k^{(i)}} \quad (2.118)$$

Algorithm 2.7 Generic Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot x_{k-1}^{(i)}, z_k)$	▷ Particle propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k x_k^{(i)}) \tilde{p}(x_k^{(i)} x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})}$	▷ Weight update
$\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$	▷ Normalisation
$\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N (w_k^{(i)})^2$	▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ then $\left\{ x_k^{(j)}, w_k^{(j)}, - \right\}_{j=1}^N \leftarrow \text{Resampling} \left(\left\{ x_k^{(i)}, w_k^{(i)} \right\}_{i=1}^N \right)$ end if	▷ Resampling

Since this measure is singular, resampling from it can only reproduce the existing particles. The regularised particle filter (RPF) [39] addresses this issue through kernel density estimation [40, 41, 42], by replacing each Dirac with a smooth kernel to obtain the absolutely continuous measure:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{(i)} K_h(x_k - x_k^{(i)}) dx_k \quad (2.119)$$

New particles are then drawn from this continuous density rather than from the discrete set, which restores diversity, as illustrated by Figure 2.4 and formalised by Definitions 2.5 and 2.6. Being absolutely continuous, this approximation also supplies the density that (2.118) lacks, and thus makes the MAP estimator of section 2.2 well-defined at the cost of a numerical maximisation.

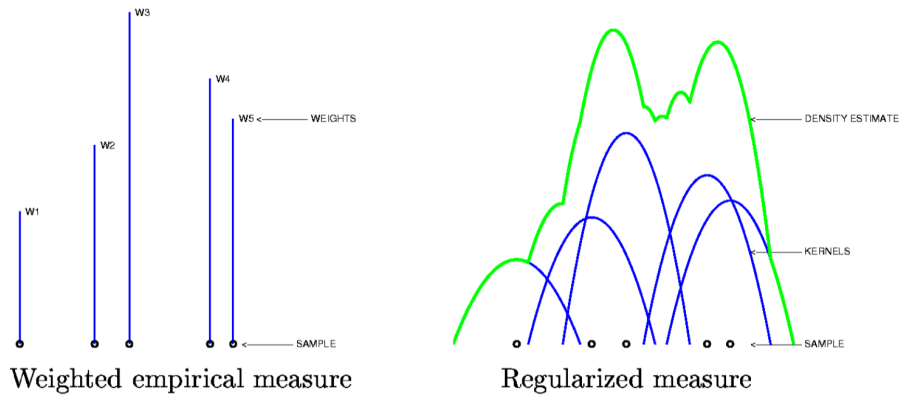


Figure 2.4: An example Monte Carlo approximation of the PDF (2.118) on the left and its kernel regularisation (2.119) on the right, for a given particle set. The latter approximation bridges the gaps between particles and spans a larger region of the state space.

Definition 2.5 (Regularisation kernels). *Let $K : \mathbb{R}^{d_x} \rightarrow \mathbb{R}$. K is a regularisation kernel if:*

$$\left\{ \begin{array}{ll} \forall x \in \mathbb{R}^{d_x}, K(x) \geq 0 & \text{positive} \\ \forall x \in \mathbb{R}^{d_x}, K(-x) = K(x) & \text{symmetric} \\ \int K(x) dx = 1 & \text{normalised} \\ \int \|x\|^2 K(x) dx < \infty & \text{of finite variance} \end{array} \right. \quad (2.120)$$

Furthermore, for K a regularisation kernel and $h > 0$, we define the rescaled kernel K_h of bandwidth h as:

$$K_h : \left\{ \begin{array}{l} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ x \mapsto \frac{1}{h^{d_x}} K\left(\frac{x}{h}\right) \end{array} \right. \quad (2.121)$$

Definition 2.6 (Regularised estimator). *Let $*$ denote the convolution operator. The regularised measure $S_{K_h}^N$ of some empirical measure S^N with respect to a regularisation kernel K_h is:*

$$S_{K_h}^N(\mu) \triangleq K_h * S^N(\mu) \quad (2.122)$$

In particular, the regularised self-normalised estimator verifies:

$$\tilde{S}_{K_h}^N(\mu, \nu) = \sum_{i=1}^N w^{(i)} (K_h * \delta_{X^{(i)}}) \quad (2.123)$$

and the corresponding estimator is:

$$\left\langle \tilde{S}_{K_h}^N(\mu, \nu), \varphi \right\rangle = \sum_{i=1}^N w^{(i)} (K_h * \varphi)(X^{(i)}) \quad (2.124)$$

by symmetry of K_h .

The choice of bandwidth h is critical to the performance of the RPF. If h is too small, K_h concentrates around the Dirac measure, and the regularisation has no practical effect on particle diversity. Conversely, the kernel density estimate over-smooths the posterior when h is too large, artificially spreading particles across regions of low probability, and degrading filter accuracy. In this section, we present two classical choices for K_h :

- **Epanechnikov kernel [43].** This kernel minimises the mean integrated squared error (MISE) among all regularisation kernels, and the associated bandwidth is optimal when the density is Gaussian with identity covariance. It is defined by:

$$\left\{ \begin{array}{l} K_{\text{Epa}} : x \mapsto \frac{d_x + 2}{2c_{d_x}} (1 - \|x\|^2) \mathbb{1}_{\|x\| \leq 1}(x) \\ h_{\text{Epa}} = \left(\frac{8(d_x + 4)(2\sqrt{\pi})^{d_x}}{c_{d_x}} \right)^{\frac{1}{d_x+4}} N^{-\frac{1}{d_x+4}} \end{array} \right. \quad (2.125)$$

where:

$$c_{d_x} \triangleq \frac{\pi^{d_x/2}}{\Gamma(d_x/2 + 1)} \quad (2.126)$$

is the volume of the $\mathbb{R}^{d_x}$ unit ball, and Γ is the Gamma function.

- **Gaussian kernel.** While suboptimal with respect to the Epanechnikov kernel, it is simpler to compute [44]. Its expression is given by:

$$\begin{cases} K_{\text{Gauss}} : x \mapsto (2\pi)^{-d_x/2} \exp\left(-\frac{\|x\|^2}{2}\right) \\ h_{\text{Gauss}} = \left(\frac{4}{d_x + 2}\right)^{\frac{1}{d_x + 4}} N^{-\frac{1}{d_x + 4}} \end{cases} \quad (2.127)$$

It must be noted that the assumptions for the Epanechnikov, and consequently the Gaussian, regularisation are not usually met in practice, since MC methods are precisely used when the PDF is not Gaussian.

However, the two conditions do not play the same role. The Gaussian hypothesis is only used to compute the optimal bandwidth, whose expression would otherwise depend on an unknown PDF. It is thus a calibration convention rather than a modelling assumption. In particular, it tends to overestimate h when the PDF is multimodal.

On the other hand, the unit covariance is a scaling requirement, and can be enforced exactly by a whitening step before the regularisation. This amounts to replacing every particle by $A_k^{-1} x_k^{(i)}$, where A_k is the Cholesky factor [45] of $\hat{P}_k$:

$$\hat{P}_k = A_k A_k^\top \quad (2.128)$$

which ensures that the covariance matrix of the new particle set is the identity. While seemingly computationally heavy, this reduces to using the following regularisation kernel on the standard particle set:

$$\tilde{K}_h : x \mapsto \frac{\det(A_k)^{-1}}{h^{d_x}} K\left(\frac{1}{h} A_k^{-1} x\right) \quad (2.129)$$

Moreover, the bandwidth h is multiplied by a modulation factor $\alpha_{\text{reg}} \in [0, 1]$ to avoid over-smoothing in cases where the PDF is multimodal. The pseudo-code for the RPF is given by Algorithm 2.8.

2.5.3 Auxiliary Particle Filter

The auxiliary particle filter (APF) [46, 47] addresses the degeneracy problem at its root by exploiting a different weakness of the SIR. Since particles are propagated before the new observation z_k is taken into account, particles that will ultimately receive low likelihood waste computational effort. The APF thus introduces a look-ahead step by using a prediction of their future likelihood to resample promising parent particles before propagation.

Formally, the APF targets the joint distribution $p(x_k, i | \mathbf{z}_{1:k})$, where $i \in \llbracket 1, N \rrbracket$ indexes the parent particle at time $k - 1$. Using Bayes' theorem and the chain rule gives:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k, i, \mathbf{z}_{1:k-1}) p(x_k | i, \mathbf{z}_{1:k-1}) p(i | \mathbf{z}_{1:k-1}) \quad (2.130)$$

Each term of this decomposition can be simplified as follows:

- Assumption 2.2 states that the current observation depends only on the current state:

$$p(z_k | x_k, i, \mathbf{z}_{1:k-1}) = p(z_k | x_k) \quad (2.131)$$

Algorithm 2.8 Regularised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, x_k^{(i)} \sim q(\cdot | x_{k-1}^{(i)}, z_k)$ ▷ Particle propagation
 $\forall i \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \frac{\tilde{p}(z_k | x_k^{(i)}) \tilde{p}(x_k^{(i)} | x_{k-1}^{(i)})}{\tilde{q}(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k})}$ ▷ Weight update
 $\forall i \in \llbracket 1, N \rrbracket, w_k^{(i)} = \tilde{w}_k^{(i)} / \sum_{j=1}^N \tilde{w}_k^{(j)}$ ▷ Normalisation
 $\hat{N}_{\text{eff}} = 1 / \sum_{i=1}^N (w_k^{(i)})^2$ ▷ Resampling criterion
if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**
 $\hat{x}_k = \sum_{i=1}^N w_k^{(i)} x_k^{(i)}$ ▷ Whitening
 $\hat{P}_k = \sum_{i=1}^N w_k^{(i)} (x_k^{(i)} - \hat{x}_k) (x_k^{(i)} - \hat{x}_k)^\top$
 $A_k \text{ s.t. } \hat{P}_k = A_k A_k^\top$
 $\{x_k^{(j)}, w_k^{(j)}, -\}_{j=1}^N \leftarrow \text{Resampling} \left(\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^N \right)$ ▷ Resampling
 $\forall j \in \llbracket 1, N \rrbracket, \varepsilon^{(j)} \sim K$ ▷ Regularisation
 $\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k \varepsilon^{(j)}$
end if

- Since the parent index uniquely determines the ancestor trajectory, conditioning on i is equivalent to conditioning on $x_{0:k-1}^{(i)}$. Moreover, Assumption 2.1 establishes that $(x_k)_{k \geq 0}$ is a Markov chain, and thus:

$$p(x_k | i, \mathbf{z}_{1:k-1}) = p(x_k | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k-1}) = p(x_k | x_{k-1}^{(i)}) \quad (2.132)$$

- Finally, assuming that the particle approximation is exact gives:

$$p(i | \mathbf{z}_{1:k-1}) \propto w_{k-1}^{(i)} \quad (2.133)$$

Combining these equations reduces (2.130) to:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k) p(x_k | x_{k-1}^{(i)}) w_{k-1}^{(i)} \quad (2.134)$$

However, $p(z_k | x_k)$ cannot be evaluated before x_k is drawn. It is thus approximated by $p(z_k | \mu_k^{(i)})$, where:

$$\mu_k^{(i)} \sim p(\cdot | x_{k-1}^{(i)}) \quad (2.135)$$

is an auxiliary particle drawn from the transition density, and used to assess the consistency of the future particle $x_k^{(i)}$ with the new observation z_k . Alternatively, $\mu_k^{(i)}$ can be chosen as the mean of this density, a deterministic choice which avoids an additional draw.

In any case, the importance density is therefore:

$$q(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | \mu_k^{(i)}) p(x_k | x_{k-1}^{(i)}) w_{k-1}^{(i)} \quad (2.136)$$

Since they are drawn from the prior distribution, (2.106) ensures that the auxiliary weights $\lambda_k^{(i)}$ follow:

$$\lambda_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k | \mu_k^{(i)}) \quad (2.137)$$

In the hope that the evolution noise is sufficiently weak for the auxiliary particles to correctly represent the future particle set, the parent indices $\{i^{(j)}\}_{j=1}^N$ are resampled according to the auxiliary weights.

The new particle set is then propagated through a standard SIS algorithm using the transition density:

$$x_k^{(j)} \sim p(\cdot | x_{k-1}^{(i^{(j)})}) \quad (2.138)$$

with their weights updating as:

$$\begin{aligned} w_k^{(j)} &\propto \frac{\tilde{p}(x_k, i^{(j)} | \mathbf{z}_{1:k})}{\tilde{q}(x_k, i^{(j)} | \mathbf{z}_{1:k})} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(j)}) \tilde{p}(x_k^{(j)} | x_{k-1}^{(i^{(j)})}) w_{k-1}^{(i^{(j)})}}{\tilde{p}(z_k | \mu_k^{(i^{(j)})}) \tilde{p}(x_k^{(j)} | x_{k-1}^{(i^{(j)})}) w_{k-1}^{(i^{(j)})}} \\ &\propto \frac{\tilde{p}(z_k | x_k^{(j)})}{\tilde{p}(z_k | \mu_k^{(i^{(j)})})} \end{aligned} \quad (2.139)$$

The full APF algorithm is given in algorithm 2.9.

Algorithm 2.9 Auxiliary Particle Filter

$\forall i \in \llbracket 1, N \rrbracket, \mu_k^{(i)} \sim p(\cdot x_{k-1}^{(i)})$	▷ Auxiliary propagation
$\forall i \in \llbracket 1, N \rrbracket, \tilde{\lambda}_k^{(i)} \propto w_{k-1}^{(i)} \tilde{p}(z_k \mu_k^{(i)})$	▷ Auxiliary update
$\forall i \in \llbracket 1, N \rrbracket, \lambda_k^{(i)} = \tilde{\lambda}_k^{(i)} / \sum_{j=1}^N \tilde{\lambda}_k^{(j)}$	▷ Normalisation
$\{-, -, i^{(j)}\}_{j=1}^N = \text{Resampling} \left(\left\{ \mu_k^{(i)}, \lambda_k^{(i)} \right\}_{i=1}^N \right)$	▷ Resampling
$\forall j \in \llbracket 1, N \rrbracket, x_k^{(j)} \sim p(\cdot x_{k-1}^{(i^{(j)})})$	▷ Particle propagation
$\forall j \in \llbracket 1, N \rrbracket, \tilde{w}_k^{(j)} \propto \frac{\tilde{p}(z_k x_k^{(j)})}{\tilde{p}(z_k \mu_k^{(i^{(j)})})}$	▷ Weight update
$\forall j \in \llbracket 1, N \rrbracket, w_k^{(j)} = \tilde{w}_k^{(j)} / \sum_{i=1}^N \tilde{w}_k^{(i)}$	▷ Normalisation

Remark 2.11 (Comparison between the RPF and the APF). *Both the RPF and the APF aim to alleviate the degeneracy and impoverishment problems of the standard PF, but they target different sources:*

- *The RPF restores diversity through kernel density estimation, addressing the severe impoverishment caused by near-deterministic dynamics. It is therefore suited to low evolution noise scenarios.*
- *The APF instead pre-selects promising parents from the incoming observation. It is preferred when the observation noise is small, since a peaked likelihood triggers an important single-step weight degeneracy.*

These preferences are rules of thumb rather than theoretical results, and the APF's pre-selection also degrades under strong evolution noise. Acting at different stages, both filters are ultimately complementary rather than exclusive.

2.5.4 Rao-Blackwellised Particle Filter

The marginalised particle filter [48], also known as the Rao-Blackwellised particle filter (RBPF) [49, 50], reduces the dimension of the state-space explored by the particle filter by exploiting a partial linear-Gaussian structure in the state-space model. Suppose that the state can be decomposed as:

$$x_k = [x_k^n, x_k^l] \quad (2.140)$$

such that (2.1) is linear and Gaussian in x_k^l conditionally on x_k^n . The navigation system can be rewritten as:

$$(S) : \begin{cases} x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \\ x_k^l = F_k^l(x_{k-1}^n) x_{k-1}^l + f_k^l(x_{k-1}^n) + w_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \quad (2.141)$$

with:

$$w_k = \begin{bmatrix} w_k^n \\ w_k^l \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} Q_k^n & (Q_k^{nl})^\top \\ Q_k^{nl} & Q_k^l \end{bmatrix}\right), \quad n_k \sim \mathcal{N}(0, R_k) \quad (2.142)$$

x_0^n of known distribution, and x_0^l Gaussian.

The idea behind the RBPF is to factorise the PDF as:

$$p(\mathbf{x}_{0:k}^l, \mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) = p(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k}) p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}) \quad (2.143)$$

where $p(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k})$ is estimated via particle filtering techniques, and $p(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k})$ is analytically solved by a Kalman filter. Here, $\mathbf{x}_{0:k}^l$ and $\mathbf{x}_{0:k}^n$ are defined similarly to $\mathbf{x}_{0:k}$.

When $Q_k^{nl} \neq 0$, the noises w_k^l and w_k^n are correlated, and the Kalman filter cannot be applied directly. Proposition 2.9 decouples them by removing the component of w_k^l correlated with w_k^n .

Proposition 2.9 (Noise decorrelation). *The decorrelated noise:*

$$\bar{w}_k^l \triangleq w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n \quad (2.144)$$

is centred, uncorrelated with w_k^n , and has covariance:

$$\bar{Q}_k^l \triangleq \mathbb{V}[\bar{w}_k^l] = Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \quad (2.145)$$

Proof. As a linear combination of the centred noises w_k^l and w_k^n , $\bar{w}_k^l$ is centred. Its covariance develops as:

$$\begin{aligned}
\bar{Q}_k^l &= \mathbb{E} [\bar{w}_k^l \bar{w}_k^{l\top}] \\
&= \mathbb{E} [(w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] \\
&= \mathbb{E} [w_k^l (w_k^l)^\top] - \mathbb{E} [w_k^l (Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] - \mathbb{E} [(Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^l)^\top] \\
&\quad + \mathbb{E} [(Q_k^{nl} (Q_k^n)^{-1} w_k^n) (Q_k^{nl} (Q_k^n)^{-1} w_k^n)^\top] \\
&= \mathbb{E} [w_k^l (w_k^l)^\top] - \mathbb{E} [w_k^l (w_k^n)^\top] (Q_k^n)^{-\top} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^l)^\top] \\
&\quad + Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^n)^\top] (Q_k^n)^{-\top} (Q_k^{nl})^\top \\
&= Q_k^l - Q_k^{nl} (Q_k^n)^{-\top} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top + Q_k^{nl} (Q_k^n)^{-1} Q_k^n (Q_k^n)^{-\top} (Q_k^{nl})^\top \\
&= Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top + Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top \\
&= Q_k^l - Q_k^{nl} (Q_k^n)^{-1} (Q_k^{nl})^\top
\end{aligned} \tag{2.146}$$

Finally, computing:

$$\begin{aligned}
\mathbb{E} [\bar{w}_k^l (w_k^n)^\top] &= \mathbb{E} [(w_k^l - Q_k^{nl} (Q_k^n)^{-1} w_k^n) (w_k^n)^\top] \\
&= \mathbb{E} [w_k^l (w_k^n)^\top] - Q_k^{nl} (Q_k^n)^{-1} \mathbb{E} [w_k^n (w_k^n)^\top] \\
&= Q_k^{nl} - Q_k^{nl} (Q_k^n)^{-1} Q_k^n \\
&= 0
\end{aligned} \tag{2.147}$$

shows that $\bar{w}_k^l$ and w_k^n are uncorrelated. $\square$

Let:

$$\bar{F}_k^l(x_{k-1}^n) \triangleq F_k^l(x_{k-1}^n) - Q_k^{nl} (Q_k^n)^{-1} F_k^n(x_{k-1}^n) \tag{2.148}$$

$$\bar{f}_k^l(x_{k-1}^n) \triangleq f_k^l(x_{k-1}^n) - Q_k^{nl} (Q_k^n)^{-1} f_k^n(x_{k-1}^n) \tag{2.149}$$

The system (2.141) can be rewritten as:

$$(\bar{S}) : \begin{cases} x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \\ x_k^l = \bar{F}_k^l(x_{k-1}^n) x_{k-1}^l + \bar{f}_k^l(x_{k-1}^n) + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \bar{w}_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases} \tag{2.150}$$

which can in turn be decomposed as:

$$\begin{aligned}
(S_1) : & \begin{cases} x_{k-1}^l = x_{k-1}^l \\ x_k^n = F_k^n(x_{k-1}^n) x_{k-1}^l + f_k^n(x_{k-1}^n) + w_k^n \end{cases} \\
(S_2) : & \begin{cases} x_k^l = \bar{F}_k^l(x_{k-1}^n) x_{k-1}^l + \bar{f}_k^l(x_{k-1}^n) + Q_k^{nl} (Q_k^n)^{-1} x_k^n + \bar{w}_k^l \\ z_k = H_k(x_k^n) x_k^l + h_k(x_k^n) + n_k \end{cases}
\end{aligned} \tag{2.151}$$

Note the similarities between each of the systems (S_1) and (S_2) , and the navigation equations under Kalman hypotheses (2.35). Indeed, (S_1) can be treated as if the propagation of the non-linear state x_k^n were an observation of x_{k-1}^l . Its absence of dynamics leads to only performing the (pre-)correction step of the Kalman filter. On the other hand, (S_2) can be handled by the usual Kalman prediction/correction.

In full generality, a standard RBPF is computed in five steps:

- **Particle propagation.** The non-linear state is propagated according to the proposal density:

$$x_k^{n(i)} \sim q\left(\cdot | x_{k-1}^{n(i)}, z_k\right) \quad (2.152)$$

- **Kalman pre-correction.** The previous linear state is corrected according to the Kalman equations applied to (S_1) :

$$\tilde{F}_k^{n(i)} = F_k^n \left(x_{k-1}^{n(i)} \right) \quad (2.153)$$

$$y_k^{n(i)} = x_k^{n(i)} - \left(\tilde{F}_k^{n(i)} \hat{x}_{k-1|k-1}^{l(i)} + f_k^n \left(x_{k-1}^{n(i)} \right) \right) \quad (2.154)$$

$$S_k^{n(i)} = \tilde{F}_k^{n(i)} P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{n(i)} \right)^\top + Q_k^n \quad (2.155)$$

$$K_k^{n(i)} = P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{n(i)} \right)^\top \left(S_k^{n(i)} \right)^{-1} \quad (2.156)$$

$$\hat{x}_{k-1|k-1}^{l(i)} \leftarrow \hat{x}_{k-1|k-1}^{l(i)} + K_k^{n(i)} y_k^{n(i)} \quad (2.157)$$

$$P_{k-1|k-1}^{l(i)} \leftarrow \left(I - K_k^{n(i)} \tilde{F}_k^{n(i)} \right) P_{k-1|k-1}^{l(i)} \quad (2.158)$$

- **Kalman prediction.** The current linear state is then predicted via the dynamics of (S_2) :

$$\tilde{F}_k^{l(i)} = \bar{F}_k^l \left(x_{k-1}^{n(i)} \right) \quad (2.159)$$

$$\hat{x}_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)} \hat{x}_{k-1|k-1}^{l(i)} + \bar{f}_k^l \left(x_{k-1}^{n(i)} \right) + Q_k^{nl} (Q_k^n)^{-1} x_k^{n(i)} \quad (2.160)$$

$$P_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)} P_{k-1|k-1}^{l(i)} \left(\tilde{F}_k^{l(i)} \right)^\top + \bar{Q}_k^l \quad (2.161)$$

$$\tilde{H}_k^{(i)} = H_k \left(x_k^{n(i)} \right) \quad (2.162)$$

$$y_k^{l(i)} = z_k - \left(\tilde{H}_k^{(i)} \hat{x}_{k|k-1}^{l(i)} + h_k \left(x_k^{n(i)} \right) \right) \quad (2.163)$$

$$S_k^{l(i)} = \tilde{H}_k^{(i)} P_{k|k-1}^{l(i)} \left(\tilde{H}_k^{(i)} \right)^\top + R_k \quad (2.164)$$

- **Weight update.** The weights corresponding to each particle are updated with respect to the new observation:

$$\tilde{w}_k^{n(i)} = w_{k-1}^{n(i)} \frac{\tilde{p} \left(z_k | x_k^{n(i)} \right) \tilde{p} \left(x_k^{n(i)} | x_{k-1}^{n(i)} \right)}{\tilde{q} \left(x_k^{n(i)} | x_{k-1}^{n(i)}, z_k \right)} \quad (2.165)$$

$$w_k^{n(i)} = \tilde{w}_k^{n(i)} \left/ \sum_{j=1}^N \tilde{w}_k^{n(j)} \right. \quad (2.166)$$

- **Kalman correction.** The linear state is corrected through the standard Kalman correction for (S_2) :

$$K_k^{l(i)} = P_{k|k-1}^{l(i)} \left(\tilde{H}_k^{(i)} \right)^\top \left(S_k^{l(i)} \right)^{-1} \quad (2.167)$$

$$\hat{x}_{k|k}^{l(i)} = \hat{x}_{k|k-1}^{l(i)} + K_k^{l(i)} y_k^{l(i)} \quad (2.168)$$

$$P_{k|k}^{l(i)} = \left(I - K_k^{l(i)} \tilde{H}_k^{(i)} \right) P_{k|k-1}^{l(i)} \quad (2.169)$$

In the RBPF, only the non-linear state is sampled, while the conditional distribution of the linear substate is propagated analytically. The particle approximation is thus the mixture [51]:

$$p(x_k | \mathbf{z}_{1:k}) dx_k \approx \sum_{i=1}^N w_k^{n(i)} \delta_{x_k^{n(i)}}(dx_k^n) \mathcal{N}(\hat{x}_{k|k}^{l(i)}, P_{k|k}^{l(i)})(dx_k^l) \quad (2.170)$$

The MMSE estimator of section 2.2 can then be approximated by:

$$\hat{x}_k = \sum_{i=1}^N w_k^{n(i)} x_k^{(i)} \quad (2.171)$$

where

$$x_k^{(i)} \triangleq \begin{bmatrix} x_k^{n(i)} \\ \hat{x}_{k|k}^{l(i)} \end{bmatrix} = \mathbb{E} \left[x_k | \mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k} \right] \quad (2.172)$$

Furthermore, since the non-linear state is deterministic when conditioned on the sampled trajectory, we have:

$$\mathbb{V} \left[x_k | \mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k} \right] = \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \quad (2.173)$$

Applying the law of total covariance finally gives the associated empirical covariance matrix:

$$\hat{P}_k = \sum_{i=1}^N w_k^{n(i)} \left(x_k^{(i)} - \hat{x}_k \right) \left(x_k^{(i)} - \hat{x}_k \right)^\top + \sum_{i=1}^N w_k^{n(i)} \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \quad (2.174)$$

where the first term is the covariance of the conditional means, and the second is the average conditional covariance.

The full RBPF algorithm is summarised in Algorithm 2.10.

Algorithm 2.10 Rao-Blackwellised Particle Filter

$\forall i \in \llbracket 1, N \rrbracket$, propagate $x_k^{n(i)}$	using (2.152)
$\forall i \in \llbracket 1, N \rrbracket$, update $\hat{x}_{k-1 k-1}^{l(i)}$ and $P_{k-1 k-1}^{l(i)}$	using (2.153) through (2.158)
$\forall i \in \llbracket 1, N \rrbracket$, predict $\hat{x}_{k k-1}^{l(i)}$ and $P_{k k-1}^{l(i)}$	using (2.159) through (2.164)
$\forall i \in \llbracket 1, N \rrbracket$, update $w_k^{n(i)}$	using (2.165) through (2.166)
$\forall i \in \llbracket 1, N \rrbracket$, compute $\hat{x}_{k k}^{l(i)}$ and $P_{k k}^{l(i)}$	using (2.167) through (2.169)

It must be noted that the version presented by Algorithm 2.10 uses a simple SIS to handle the particles, but any more advanced algorithm presented in this section can be used instead. Furthermore, the likelihood $\tilde{p}(z_k | x_k^{n(i)})$ and the transition $\tilde{p}(x_k^{n(i)} | x_{k-1}^{n(i)})$ in the weight update are white Gaussian densities, with covariances $S_k^{l(i)}$ and $S_k^{n(i)}$ respectively. This is why $S_k^{l(i)}$ is computed within the Kalman prediction step, before the weight update. However, the correction step itself must be performed after the weight update in situations where a resampling step is involved.

Remark 2.12 (Single Kalman filter). *In general, one Kalman filter is required per particle, since $\tilde{F}_k^{n(i)}$, $\tilde{F}_k^{l(i)}$ and $\tilde{H}_k^{(i)}$ all depend on $x_{k-1}^{n(i)}$ and $x_k^{n(i)}$. However, if these matrices do not depend on the non-linear state, then the gains $K_k^{n(i)}$ and $K_k^{l(i)}$, and the covariances $P_{k|k}^{l(i)}$ are identical for all particles. In this scenario, a single Kalman filter can be used for the whole particle set, which substantially reduces the computational cost of the RBPF.*

Remark 2.13 (Triangular model). *If (2.141) is such that $F_k^n = 0$, the system (S) is called triangular. In this scenario, x_k^n does not depend on x_{k-1}^l , which decouples the two filters entirely. The Kalman pre-correction step can thus be omitted, and the RBPF becomes a simple interleaving between a PF and a collection of Kalman filters.*

CHAPTER 3

GAUSSIAN PROCESSES

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3.1 Introduction

The analytical and Monte Carlo methods of chapter 2 all assume the observation function h_k to be fully known. However, many navigation scenarios depend on a physical field that is either unavailable or known only through scarce and noisy measurements.

A first way to fill this gap is to reconstruct the field directly from the measurements:

- deterministic interpolation, through nearest-neighbour selection, multilinear interpolation over a regular grid, or spline fitting;
- linear regression on a fixed basis, such as polynomials or sinusoids;
- more expressive models, such as neural networks.

All of them return a single value at the queried location, without quantifying how much it can be trusted. Their use in the filters of chapter 2, however, requires the observation likelihood $p(z_k|x_k)$, a full distribution rather than a point prediction. The reconstruction error must then be absorbed into the observation noise, usually as a covariance held constant over the whole domain, although this error is almost zero next to a measurement and grows in sparsely sampled regions.

This chapter instead models such fields as a Gaussian process (GP), an adaptable framework that encodes both a prior expectation of the field and its spatial correlation structure. Together with its uncertainty, the resulting estimator can then serve as a surrogate observation model [52, 53] for the filters of chapter 2.

The rest of this chapter is organised as follows:

- Section 3.2 defines GPs through their mean functions and covariance kernels. It discusses the choice of kernel and the estimation of its hyperparameters.
- Section 3.3 then introduces kriging as a map reconstruction tool in situations where we are given preexisting data samples.
- Section 3.4 addresses the cost of inverting large Gram matrices in kriging by approximating k with a reduced-rank kernel, therefore lowering the effective complexity of kriging methods.
- Finally, section 3.5 derives two representations of the mean function over a dictionary of functions, used to reconstruct the field when no prior knowledge is available.

3.2 Definition and Properties

Formally, a stochastic process generalises the notion of random variable to a random function, *i.e.* a family $(f(x))_{x \in \mathcal{U}}$ of random variables indexed by a set $\mathcal{U}$, which may represent time $\mathbb{R}_+$, physical space $\mathbb{R}^2$ or $\mathbb{R}^3$, or any other domain. As shown by Definition 3.1, a GP is a special case of stochastic process [54].

Definition 3.1 (Gaussian process). *For $d_z \in \mathbb{N}^*$, a stochastic process $(f(x))_{x \in \mathcal{U}}$ taking values in $\mathbb{R}^{d_z}$ is a GP if every finite linear combination of its values follows a Gaussian distribution. Formally, for all $n \in \mathbb{N}^*$, all $x_1, \dots, x_n \in \mathcal{U}$ and all $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, there exist $m \in \mathbb{R}^{d_z}$ and $P \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$ such that $P \succeq 0$ and:*

$$\sum_{i=1}^n a_i^\top f(x_i) \sim \mathcal{N}(m, P) \quad (3.1)$$

The GP is said to be non-degenerate if $P \succ 0$ whenever the points $x_1, \dots, x_n$ are pairwise distinct and the coefficients $a_1, \dots, a_n$ are not all zero.

A GP is associated with a mean function:

$$m : \begin{cases} \mathcal{U} \rightarrow \mathbb{R}^{d_z} \\ x \mapsto \mathbb{E}[f(x)] \end{cases} \quad (3.2)$$

and a covariance kernel:

$$k : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ (x, x') \mapsto \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))^\top] \end{cases} \quad (3.3)$$

3.2.1 Basic Properties

As shown by Proposition 3.1, m and k entirely characterise every finite-dimensional marginal of f . By Kolmogorov's extension theorem, they also uniquely determine the law of f . We thus write $f \sim \mathcal{GP}(m, k)$ for a GP with mean function m and covariance kernel k .

Proposition 3.1 (Finite-dimensional marginals). *For $n \in \mathbb{N}^*$, let $x_1, \dots, x_n$ be pairwise distinct points of $\mathcal{U}$. The joint distribution of $F \triangleq [f(x_1)^\top, \dots, f(x_n)^\top]^\top$ is the multivariate Gaussian:*

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \right) \quad (3.4)$$

where each block $k(x_i, x_j) \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$.

Proof. According to Definition 3.1, every linear combination:

$$a^\top F = \sum_{i=1}^n a_i^\top f(x_i), \quad a \triangleq \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^{nd_z} \quad (3.5)$$

is Gaussian. By the Cramér-Wold theorem [55] F is thus also a Gaussian vector of $\mathbb{R}^{nd_z}$.

Such a vector is entirely determined by its mean and covariance, which we compute. By linearity of the expectation, we have:

$$\mathbb{E}[F] = \begin{bmatrix} \mathbb{E}[f(x_1)] \\ \vdots \\ \mathbb{E}[f(x_n)] \end{bmatrix} = \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \quad (3.6)$$

By definition of k , it comes that:

$$\begin{aligned} \mathbb{V}[F] &= \begin{bmatrix} \mathbb{E} [(f(x_i) - m(x_i))(f(x_j) - m(x_j))^\top] \end{bmatrix} \\ &= \begin{bmatrix} k(x_i, x_j) \end{bmatrix} \end{aligned} \quad (3.7)$$

which is exactly (3.4). $\square$

For two tuples of inputs $\mathbf{x} \triangleq (x_1, \dots, x_n) \in \mathcal{U}^n$ and $\mathbf{x}' \triangleq (x'_1, \dots, x'_p) \in \mathcal{U}^p$ of sizes $n, p \in \mathbb{N}^*$, we use a vectorised notation to write:

$$f(\mathbf{x}) \triangleq \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \quad (3.8)$$

$$m(\mathbf{x}) \triangleq \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \quad (3.9)$$

$$k(\mathbf{x}, \mathbf{x}') \triangleq \begin{bmatrix} k(x_1, x'_1) & \cdots & k(x_1, x'_p) \\ \vdots & \ddots & \vdots \\ k(x_n, x'_1) & \cdots & k(x_n, x'_p) \end{bmatrix} \quad (3.10)$$

Using these notations, Proposition 3.1 reads compactly:

$$\forall n \in \mathbb{N}^*, \forall \mathbf{x} \in \mathcal{U}^n \text{ pairwise distinct, } f(\mathbf{x}) \sim \mathcal{N}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x})) \quad (3.11)$$

A GP is thus a consistent generalisation of the multivariate Gaussian distribution to an infinite index set. Both of its parameters admit a direct interpretation. The mean function m encodes prior knowledge about the average behaviour of f , while the kernel k encodes the correlation structure. The diagonal blocks $k(x, x)$ measure the variance of f at x , and the off-diagonal blocks $k(x, x')$ for $x \neq x'$ capture how the values at x and x' co-vary. In the context of this thesis, m is interpreted as a simplified model of the observation function, and k as a measure of the uncertainty and spatial variability around this model.

The estimation of the mean function is deferred to section 3.5. Instead, the remainder of this section focuses on the choice of covariance kernel and, in particular, the estimation of its hyperparameters.

3.2.2 Covariance Kernels

While the mean function generally reflects application-specific prior knowledge, it is the covariance kernel that governs the smoothness and correlation structure of the GP, and thus the class of functions it can represent.

For the GP to be non-degenerate, every matrix $k(\mathbf{x}, \mathbf{x})$ at distinct points must be SPD in the sense of Definition 2.1. Definition 3.2 recasts this as a condition on k alone [56], and Proposition 3.2 proves the equivalence between the two.

Definition 3.2 (Kernel symmetric positive definiteness). A kernel $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is symmetric positive semi-definite (SPSD) if it is:

- **Symmetric:**

$$\forall x, x' \in \mathcal{U}, k(x', x) = k(x, x')^\top \quad (3.12)$$

- **Positive semi-definite:**

$$\begin{aligned} \forall n \in \mathbb{N}^*, \forall x_1, \dots, x_n \in \mathcal{U}, \forall a_1, \dots, a_n \in \mathbb{R}^{d_z}, \\ \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j \geq 0 \end{aligned} \quad (3.13)$$

If the inequality is strict for every choice of pairwise distinct points $x_1, \dots, x_n$ and coefficients $a_1, \dots, a_n$ not all zero, then k is symmetric positive definite (SPD).

Proposition 3.2 (Kernel/matrix definiteness equivalence). The following equivalences hold:

- $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is a SPSPD kernel if and only if, for every $n \in \mathbb{N}^*$ and every $\mathbf{x} \in \mathcal{U}^n$, $k(\mathbf{x}, \mathbf{x}) \in \mathcal{M}_{nd_z, nd_z}(\mathbb{R})$ is a SPSPD matrix.
- k is a SPD kernel if and only if, for every such tuple $\mathbf{x}$ of pairwise distinct points, $k(\mathbf{x}, \mathbf{x})$ is a SPD matrix.

Proof. Forward implication. Let $n \in \mathbb{N}^*$ and $\mathbf{x} \in \mathcal{U}^n$. It comes from the symmetry of k that:

$$\begin{aligned} k(\mathbf{x}, \mathbf{x}) &= \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \\ &= \begin{bmatrix} k(x_1, x_1)^\top & \cdots & k(x_n, x_1)^\top \\ \vdots & \ddots & \vdots \\ k(x_1, x_n)^\top & \cdots & k(x_n, x_n)^\top \end{bmatrix} \\ &= \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix}^\top \\ &= k(\mathbf{x}, \mathbf{x})^\top \end{aligned} \quad (3.14)$$

Let $a \in \mathbb{R}^{nd_z}$. We decompose $a = [a_1^\top, \dots, a_n^\top]^\top$ with $a_i \in \mathbb{R}^{d_z}$, $i \in [1, n]$ to obtain:

$$\begin{aligned} a^\top k(\mathbf{x}, \mathbf{x}) a &= [a_1^\top \cdots a_n^\top] \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j \\ &\geq 0 \end{aligned} \quad (3.15)$$

Reverse implication. Let $x, x' \in \mathcal{U}$. By symmetry of $k((x, x'), (x, x'))$, it comes that:

$$\begin{aligned} \begin{bmatrix} k(x, x) & k(x, x') \\ k(x', x) & k(x', x') \end{bmatrix} &= \begin{bmatrix} k(x, x) & k(x, x') \\ k(x', x) & k(x', x') \end{bmatrix}^\top \\ &= \begin{bmatrix} k(x, x)^\top & k(x', x)^\top \\ k(x, x')^\top & k(x', x')^\top \end{bmatrix} \end{aligned} \quad (3.16)$$

In particular, the upper-right block gives:

$$k(x, x') = k(x', x)^\top \quad (3.17)$$

Let $n \in \mathbb{N}^*$, $\mathbf{x} \in \mathcal{U}^n$, and $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, and set $a \triangleq [a_1^\top, \dots, a_n^\top]^\top \in \mathbb{R}^{nd_z}$. Then:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i, x_j) a_j &= [a_1^\top \ \cdots \ a_n^\top] \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\ &= a^\top k(\mathbf{x}, \mathbf{x}) a \\ &\geq 0 \end{aligned} \quad (3.18)$$

Strict version. The same computations hold with strict inequalities, provided the inputs are pairwise distinct and the coefficients not all zero. $\square$

Remark 3.1 (Necessity of distinct points). *While pairwise distinctness does not intervene in the computation of the strict version above, it is necessary for the SPD condition to be satisfiable. Indeed, at coinciding inputs $x_1 = x_2$, the choice $a_1 = -a_2 = 1$, with the other coefficients set to zero, cancels the quadratic form for non-zero coefficients, so no kernel could be SPD.*

Remark 3.2 (Non-diagonal blocks). *While Proposition 3.2 equates the SPDness of k with that of $k(\mathbf{x}, \mathbf{x})$ at distinct points, it does not make the off-diagonal blocks $k(x, x')$ SPD themselves. For $x \neq x'$, these may be non-positive or even non-symmetric.*

Throughout the rest of this document, we assume that the covariance kernel k is SPD and that the points of every collection $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{U}^n$ are pairwise distinct. This ensures that we only work with non-degenerate GPs.

Common Kernels

Assume in this section that $\mathcal{U}$ is a finite-dimensional normed vector space, *i.e.* $\mathcal{U} = \mathbb{R}^{d_x}$ up to isomorphism. A natural way to build kernels on $\mathcal{U}$ is to let the correlation between the field values at two points x, x' depend only on the distance $\|x - x'\|$ between them. Kernels of this form are called radial kernels [57], and are formally defined by Definition 3.3.

Since f typically represents a physical field, observations tend to become less correlated as the distance between them increases. Definition 3.4 introduces a common family of radial kernels modelling this behaviour, called the Matérn kernels [58]. Proposition 3.3 then establishes their continuity and SPDness.

Definition 3.3 (Radial kernel). *A kernel $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is radial if there exists a function $f : \mathbb{R}_+ \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R})$ and a norm $\|\cdot\|$ on $\mathbb{R}^{d_x}$ such that:*

$$\forall x, x' \in \mathbb{R}^{d_x}, k(x, x') = f(\|x - x'\|) \quad (3.19)$$

To simplify notations, we identify f with k in the remainder of this document, such that we write:

$$\forall x, x' \in \mathbb{R}^{d_x}, k(x, x') = k(\|x - x'\|) \quad (3.20)$$

Definition 3.4 (Matérn kernel). *Assume that $d_z = 1$, and let $\nu, l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel of smoothness ν , correlation length l and characteristic variance σ_f^2 is the radial kernel defined as:*

$$k_\nu : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{l} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{l} \right) \end{cases} \quad (3.21)$$

where Γ is the Gamma function, and K_ν is the modified Bessel function of the second kind [59].

Proposition 3.3 (Properties of Matérn kernels). *Let $\nu, l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel associated with ν , l and σ_f^2 is a continuous SPD radial kernel.*

Proof. Let k_ν denote such a function. We also let:

$$f : \begin{cases} \mathbb{R}_+ \rightarrow \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} r^\nu K_\nu(r) \end{cases} \quad \text{and} \quad \tilde{r} : \begin{cases} \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \rightarrow \mathbb{R}_+ \\ x, x' \mapsto \frac{\sqrt{2\nu}}{l} \|x - x'\| \end{cases} \quad (3.22)$$

$\tilde{r}$ and K_ν are continuous on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$ and $\mathbb{R}_+^*$ respectively, and the latter admits the following asymptotic expansion as $r \rightarrow 0^+$:

$$K_\nu(r) = 2^{\nu-1} \Gamma(\nu) r^{-\nu} + o(r^{-\nu}) \quad (3.23)$$

Therefore, f can be continuously extended to 0 by setting $f(0) = \sigma_f^2$, and k_ν is thus continuous on the entirety of $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$.

Let $x, x' \in \mathbb{R}^{d_x}$. By symmetry of the norm, $\tilde{r}(x, x') = \tilde{r}(x', x)$, and thus:

$$k_\nu = f \circ \tilde{r} \quad (3.24)$$

is symmetric, with $\circ$ being the composition operator.

Furthermore, Bochner's theorem [60] states that a continuous integrable function with a strictly positive Fourier transform is SPD. It can be shown [61] that the Fourier transform of k_ν on $\mathbb{R}^{d_x}$ is:

$$S : \begin{cases} \mathbb{R}^{d_x} \rightarrow \mathbb{R} \\ \omega \mapsto C \left(\frac{2\nu}{l^2} + \|\omega\|^2 \right)^{-(\nu+d_x/2)} \end{cases} \quad (3.25)$$

where $C > 0$ is a normalising constant which only depends on d_x and the parameters of k_ν . Since $\nu > 0$, $l > 0$, it comes that:

$$\forall \omega \in \mathbb{R}^{d_x}, S(\omega) > 0 \quad (3.26)$$

and thus k_ν is a SPD kernel. By definition, it is also a radial kernel. $\square$

The parameter ν controls the regularity. Indeed, for $p \in \mathbb{N}$, it can be shown [62] that a GP equipped with k_ν is p times mean-square differentiable if and only if $\nu > p$. For half-integer $\nu = p + \frac{1}{2}$, Matérn kernels can be written as the product of an exponential and a polynomial function of degree p :

$$k_{p+1/2}(r) = \sigma_f^2 \exp\left(-\frac{\sqrt{2\nu}r}{l}\right) \frac{\Gamma(p+1)}{\Gamma(2p+1)} \sum_{i=0}^p \frac{(p+i)!}{i!(p-i)!} \left(\frac{2\sqrt{2\nu}r}{l}\right)^{p-i} \quad (3.27)$$

In particular, for $p \in \llbracket 0, 2 \rrbracket$, we have:

- **Exponential kernel.**

$$k_{1/2}(r) = \sigma_f^2 \exp\left(-\frac{r}{l}\right) \quad (3.28)$$

- **Once-differentiable Matérn kernel.**

$$k_{3/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3}r}{l}\right) \exp\left(-\frac{\sqrt{3}r}{l}\right) \quad (3.29)$$

- **Twice-differentiable Matérn kernel.**

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{l} + \frac{5r^2}{3l^2}\right) \exp\left(-\frac{\sqrt{5}r}{l}\right) \quad (3.30)$$

As $\nu \rightarrow +\infty$, the Matérn kernel converges to the infinitely differentiable squared exponential (SE) kernel:

$$k_{SE}(r) \triangleq k_\infty(r) = \sigma_f^2 \exp\left(-\frac{r^2}{2l^2}\right) \quad (3.31)$$

Figure 3.1 illustrates the radial decay of Matérn covariance kernels for different smoothness parameters $\nu \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$, and their limit as $\nu \rightarrow +\infty$. The kernels are evaluated with unit variance $\sigma_f^2 = 1$ and correlation length $l = 1$ to facilitate comparison.

Remark 3.3 (Anisotropic kernels). *The norm used in Definition 3.3 does not necessarily need to be the canonical norm of $\mathbb{R}^{d_x}$. In particular, for $M \succ 0$, the Mahalanobis norm:*

$$x \in \mathbb{R}^{d_x} \mapsto \|x\| \triangleq \sqrt{x^\top M x} \quad (3.32)$$

may produce anisotropic radial kernels in situations where M is not proportional to the identity matrix.

Remark 3.4 (Multi-scale kernels). *It can be easily verified that a linear combination of SPD kernels with positive weights is itself a SPD kernel. In particular, for $n \in \mathbb{N}^*$, $\sigma_1^2, \dots, \sigma_n^2 \in \mathbb{R}_+^*$ and $k^{(1)}, \dots, k^{(n)}$ Matérn kernels of correlation lengths $l_1 < \dots < l_n \in \mathbb{R}_+^*$, the multi-scale kernel:*

$$k_{MS}(r) \triangleq \sum_{i=1}^n \sigma_i^2 k^{(i)}(r) \quad (3.33)$$

is a valid SPD kernel. When the correlation lengths have different orders of magnitude, such kernels can be used to capture local fluctuations and global trends simultaneously.

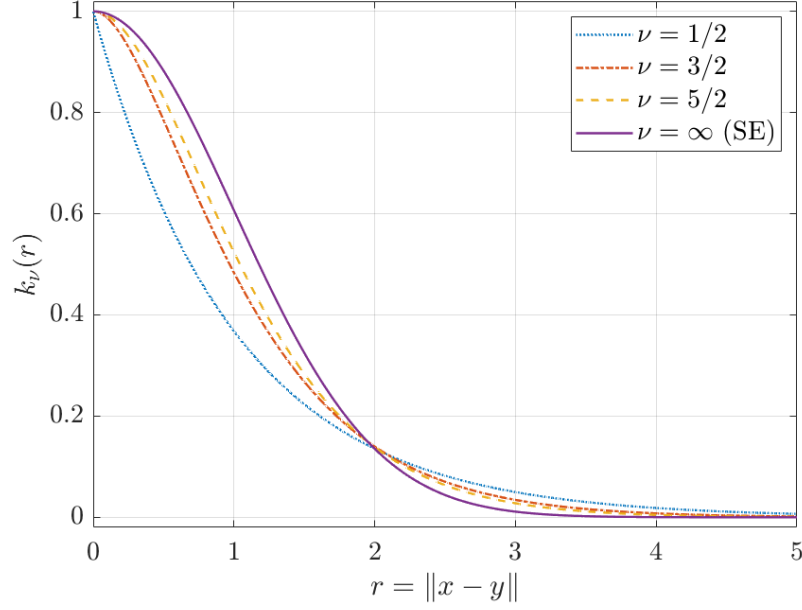


Figure 3.1: Radial profile of Matérn kernels with increasing smoothness, together with their squared exponential limit. Larger values of ν produce smoother kernels, with flatter behaviours near the origin and lighter tails.

3.2.3 Hyperparameter Estimation

TODO: à reprendre

The kernel depends on hyperparameters θ , such as σ_f^2 and l for the SE kernel, that must be inferred from the data. Given training inputs X and the corresponding observations Z , the standard approach maximises the marginal likelihood of the observations, a procedure known as evidence maximisation [63]. Writing $K_\theta \triangleq k_\theta(X, X)$ for the kernel matrix and assuming, in the absence of prior information, a zero mean function, the observations are jointly Gaussian:

$$p(Z|X, \theta) = \frac{\exp\left(-\frac{1}{2}Z^\top K_\theta^{-1}Z\right)}{\sqrt{(2\pi)^n \det(K_\theta)}} \quad (3.34)$$

Proposition 3.4 (Maximum likelihood hyperparameters). *A hyperparameter vector θ maximising (3.34) satisfies, for each component θ_i :*

$$\text{tr} \left[((K_\theta^{-1}Z)(K_\theta^{-1}Z)^\top - K_\theta^{-1}) \frac{\partial K_\theta}{\partial \theta_i} \right] = 0 \quad (3.35)$$

Proof. Maximising (3.34) is equivalent to maximising its logarithm:

$$\ell(\theta) \triangleq \log p(Z|X, \theta) = -\frac{1}{2} (Z^\top K_\theta^{-1}Z + \log \det(K_\theta)) - \frac{n}{2} \log(2\pi) \quad (3.36)$$

A maximiser cancels every partial derivative. Using the differential of the matrix inverse, $\frac{\partial}{\partial \theta_i} K_\theta^{-1} = -K_\theta^{-1} \frac{\partial K_\theta}{\partial \theta_i} K_\theta^{-1}$, and Jacobi's formula, $\frac{\partial}{\partial \theta_i} \log \det(K_\theta) =$

$\text{tr} \left(K_\theta^{-1} \frac{\partial K_\theta}{\partial \theta_i} \right)$, it comes that:

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_i} = 0 &\iff Z^\top K_\theta^{-1} \frac{\partial K_\theta}{\partial \theta_i} K_\theta^{-1} Z - \text{tr} \left(K_\theta^{-1} \frac{\partial K_\theta}{\partial \theta_i} \right) = 0 \\ &\iff \text{tr} \left[((K_\theta^{-1} Z)(K_\theta^{-1} Z)^\top - K_\theta^{-1}) \frac{\partial K_\theta}{\partial \theta_i} \right] = 0 \end{aligned} \quad (3.37)$$

using the cyclic property of the trace to rewrite the scalar quadratic form $Z^\top K_\theta^{-1} \frac{\partial K_\theta}{\partial \theta_i} K_\theta^{-1} Z$ as $\text{tr} \left((K_\theta^{-1} Z)(K_\theta^{-1} Z)^\top \frac{\partial K_\theta}{\partial \theta_i} \right)$, since K_θ^{-1} is symmetric. $\square$

Remark 3.5 (Non-convexity). *The log-marginal-likelihood is generally non-convex in θ , so (3.35) is solved numerically by gradient-based optimisation [64], and may admit several local maxima corresponding to different interpretations of the data, e.g. a low-noise rapidly-varying field versus a high-noise slowly-varying one.*

3.3 Kriging

Gaussian process regression, also known as kriging [65, 66, 67], estimates the value of a field at an unobserved location using noisy measurements of the same field at known locations. Formally, let $f \sim \mathcal{GP}(m, k) : \mathcal{U} \rightarrow \mathbb{R}^{d_z}$, and $x \in \mathcal{U}$. We wish to estimate the law of:

$$z \triangleq f(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R), \quad R \succeq 0 \quad (3.38)$$

Without further information regarding f , we would only be able to state that z follows a Gaussian distribution of expectation $m(x)$ and covariance $k(x, x) + R$. However, we are given $\tilde{n} \in \mathbb{N}^*$ distinct training points $\tilde{\mathbf{x}} \triangleq [\tilde{x}_1, \dots, \tilde{x}_{\tilde{n}}]^\top$ and their corresponding noisy observations:

$$\tilde{\mathbf{z}} \triangleq \begin{bmatrix} f(\tilde{x}_1) + \varepsilon_1 \\ \vdots \\ f(\tilde{x}_{\tilde{n}}) + \varepsilon_{\tilde{n}} \end{bmatrix}, \quad \varepsilon_i \sim \mathcal{N}(0, \tilde{R}), \quad \tilde{R} \succeq 0 \quad (3.39)$$

where the observation noises are also independent of f .

The goal of this section is to derive the best linear unbiased estimator (BLUE) of z with respect to $\tilde{\mathbf{z}}$, *i.e.* the estimator:

$$\hat{z} = W^\top \tilde{\mathbf{z}} \quad (3.40)$$

parametrised by a weight matrix:

$$W = \begin{bmatrix} w_1 \\ \vdots \\ w_{\tilde{n}} \end{bmatrix} \in \mathcal{M}_{\tilde{n}d_z, d_z}(\mathbb{R}) \quad (3.41)$$

such that the error vector has an expectation of 0 and a minimum covariance.

Since $\hat{z} - z$ is a vector, its covariance is a matrix, and cannot be minimised as such. To alleviate this issue, we minimise its trace, which is exactly the total mean error:

$$\text{tr}(\mathbb{V}[\hat{z} - z]) = \mathbb{E} [\|\hat{z} - z\|^2] \quad (3.42)$$

In the following, we present three kriging estimates, based on increasingly relaxed assumptions made on our knowledge of the underlying GP. Each method relies on the inversion of the Gram matrix:

$$\begin{aligned} G(\tilde{\mathbf{x}}) &\triangleq k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{n}} \otimes \tilde{R} \\ &= \begin{bmatrix} k(\tilde{x}_1, \tilde{x}_1) + \tilde{R} & \cdots & k(\tilde{x}_1, \tilde{x}_{\tilde{n}}) \\ \vdots & \ddots & \vdots \\ k(\tilde{x}_{\tilde{n}}, \tilde{x}_1) & \cdots & k(\tilde{x}_{\tilde{n}}, \tilde{x}_{\tilde{n}}) + \tilde{R} \end{bmatrix} \end{aligned} \quad (3.43)$$

of $\tilde{\mathbf{x}}$ with respect to f , where $\otimes$ is the Kronecker matrix product. Since k is SPD and the training points are distinct, $G(\tilde{\mathbf{x}})$ is SPD, and thus invertible, even in the noiseless case $\tilde{R} = 0$.

We further assume that the query point x differs from the training points $\tilde{\mathbf{x}}$, so that the joint law of $(z, \tilde{\mathbf{z}})$ is non-degenerate. This hypothesis is not restrictive and is only needed for the proofs. Indeed, the estimators are continuous in x and thus remain valid at the training points as soon as m and k are continuous, which we assume.

3.3.1 Simple Kriging

Simple kriging is the simplest form of kriging, and assumes that the mean function m is known. Proposition 3.5 gives a closed-form expression of the simple kriging estimator and its associated covariance matrix, while Proposition 3.7 shows that it coincides with the conditional distribution of z given $\tilde{\mathbf{z}}$.

Proposition 3.5 (Simple kriging estimator). *Using the notations introduced in section 3.3, the simple kriging estimator of z and its error covariance are:*

$$\begin{cases} \hat{z}_s(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = m(x) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\ \hat{R}_s(x, \tilde{\mathbf{x}}) = k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (3.44)$$

Proof. Since m is known, we may assume it to be equal to 0 by working with $f - m$ instead of f , without any loss of generality. In the centred case, (3.42) can be developed as follows:

$$\mathbb{V}[W^\top \tilde{\mathbf{z}} - z] = W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \quad (3.45)$$

By symmetry of $G(\tilde{\mathbf{x}})$, differentiating (3.42) with respect to W yields:

$$\begin{aligned} &\frac{\partial}{\partial W} (\text{tr} (W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R)) = 0 \\ \iff &\frac{\partial}{\partial W} (\text{tr} (W^\top G(\tilde{\mathbf{x}}) W)) - 2 \frac{\partial}{\partial W} (\text{tr} (W^\top k(\tilde{\mathbf{x}}, x))) = 0 \\ \iff &2G(\tilde{\mathbf{x}}) W - 2k(\tilde{\mathbf{x}}, x) = 0 \\ \iff &W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (3.46)$$

The simple kriging estimator is thus:

$$\hat{z}_s = W^\top \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \quad (3.47)$$

and its corresponding covariance:

$$\begin{aligned}
\hat{R}_s &= W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, x) + R \\
&= k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} G(\tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\
&\quad - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) + k(x, x) + R \\
&= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)
\end{aligned} \tag{3.48}$$

The general case is obtained by adding m back into f . $\square$

Remark 3.6 (Error covariance minimisation). *While the simple kriging weights $W = G^{-1}k(\tilde{\mathbf{x}}, x)$ were obtained by minimising the trace of $\mathbb{V}[\hat{z} - z]$, they actually minimise the whole covariance in the SPSPD sense. Indeed:*

$$\begin{aligned}
\mathbb{V}[\hat{z} - z] - \hat{R}_s &= W^\top G(\tilde{\mathbf{x}}) W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}}) W + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\
&= (W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x))^\top G(\tilde{\mathbf{x}}) (W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)) \\
&\succeq 0
\end{aligned} \tag{3.49}$$

with equality if and only if $W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)$ since $G(\tilde{\mathbf{x}}) \succ 0$.

Proposition 3.6 (Exact interpolation in the noiseless case). *Assume that $\tilde{R} = 0$, and let $i \in \llbracket 1, \tilde{n} \rrbracket$. The simple kriging estimator interpolates the data at $\tilde{x}_i$ exactly:*

$$\begin{cases} \hat{z}_s(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = \tilde{z}_i \\ \hat{R}_s(\tilde{x}_i, \tilde{\mathbf{x}}) = R \end{cases} \tag{3.50}$$

In particular, the error covariance reduces to the query noise R alone, and vanishes when $R = 0$. On the contrary, a positive training noise $\tilde{R} \succ 0$ turns kriging into a smoother rather than an interpolator [68].

Proof. Since $\tilde{R} = 0$, the Gram matrix reduces to $G(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})$. Let $e_i \triangleq [0, \dots, 0, 1, 0, \dots, 0]^\top$ be the i -th canonical basis vector of $\mathbb{R}^{\tilde{n}}$. The matrix $e_i^\top \otimes I_{d_z}$ extracts the i -th block of a stacked matrix, such that:

$$(e_i^\top \otimes I_{d_z}) G(\tilde{\mathbf{x}}) = (e_i^\top \otimes I_{d_z}) k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) = k(\tilde{x}_i, \tilde{\mathbf{x}}) \tag{3.51}$$

The kriging weights thus reduce to this selector:

$$k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} = (e_i^\top \otimes I_{d_z}) G(\tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} = e_i^\top \otimes I_{d_z} \tag{3.52}$$

Evaluating (3.44) at $x = \tilde{x}_i$ by the continuity yields:

$$\begin{aligned}
\hat{z}_s(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) &= m(\tilde{x}_i) + k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\
&= m(\tilde{x}_i) + (e_i^\top \otimes I_{d_z}) (\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})) \\
&= m(\tilde{x}_i) + \tilde{z}_i - m(\tilde{x}_i) \\
&= \tilde{z}_i
\end{aligned} \tag{3.53}$$

Similarly:

$$\begin{aligned}
\hat{R}_s(\tilde{x}_i, \tilde{\mathbf{x}}) &= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, \tilde{x}_i) \\
&= k(\tilde{x}_i, \tilde{x}_i) + R - (e_i^\top \otimes I_{d_z}) k(\tilde{\mathbf{x}}, \tilde{x}_i) \\
&= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{x}_i) \\
&= R
\end{aligned} \tag{3.54}$$

$\square$

Proposition 3.7 (Gaussian conditioning interpretation). *The conditional distribution of z knowing $\tilde{\mathbf{z}}$ is Gaussian with mean $\hat{z}_s$ and covariance $\hat{R}_s$:*

$$z|\tilde{\mathbf{z}} \sim \mathcal{N}\left(\hat{z}_s(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m), \hat{R}_s(x, \tilde{\mathbf{x}})\right) \quad (3.55)$$

Proof. Since they are evaluations of the same GP under an additive Gaussian white noise, z and $\tilde{\mathbf{z}}$ are jointly Gaussian. Their joint distribution is:

$$\begin{bmatrix} z \\ \tilde{\mathbf{z}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} m(x) \\ m(\tilde{\mathbf{x}}) \end{bmatrix}, \begin{bmatrix} k(x, x) + R & k(x, \tilde{\mathbf{x}}) \\ k(\tilde{\mathbf{x}}, x) & G(\tilde{\mathbf{x}}) \end{bmatrix}\right) \quad (3.56)$$

As noted at the start of this section, this joint covariance is SPD. The Gaussian conditioning formula (see Proposition A.1) can thus be used to give the desired result. $\square$

This result is a special case of the more general result established in Proposition 2.1. Since the unbiased hypothesis of the BLUE is automatically verified by the simple kriging estimator, the minimisation criterion (3.42) amounts to finding the MMSE estimator of z given $\tilde{\mathbf{z}}$, which is always the conditional expectation.

Remark 3.7 (Simultaneous estimation). *The simultaneous estimation of f at several query points $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{U}^n$ can be done by directly replacing x with $\mathbf{x}$ in (3.44). In particular, the $\mathcal{O}(\tilde{n}^3)$ cost of inverting $G(\tilde{\mathbf{x}})$ can be shared across the n estimations whether they are performed successively or simultaneously.*

However, computing only the n diagonal blocks of $\hat{R}_s(\mathbf{x}, \tilde{\mathbf{x}})$ costs $\mathcal{O}(\tilde{n}^3 + n\tilde{n}^2)$, whereas forming the full matrix before discarding its off-diagonal blocks costs $\mathcal{O}(\tilde{n}^3 + n\tilde{n}^2 + n^2\tilde{n})$ due to the extra off-diagonal matrix products. One must therefore be careful to only compute the block diagonal of $\hat{R}_s(\mathbf{x}, \tilde{\mathbf{x}})$ to avoid unnecessary calculations.

3.3.2 Ordinary Kriging

Assuming the mean function to be known is a very strong assumption, that is often not verified in real case scenarios. While an approximated model could be used instead of the real m , this would cause the kriging estimator to be over-confident in its estimation. The ordinary kriging extends its simple counterpart by assuming that m is an unknown constant $m \in \mathbb{R}^{d_z}$, and learning m alongside the weights.

The unbiased assumption of the ordinary kriging estimator yields:

$$\begin{aligned} & \mathbb{E}[\hat{z} - z] = 0 \\ \iff & W^\top \mathbb{E}[\tilde{\mathbf{z}}] - \mathbb{E}[z] = 0 \\ \iff & \left(\sum_{i=1}^{\tilde{n}} w_i^\top - I_{d_z} \right) m = 0 \\ \iff & C^\top W - I_{d_z} = 0 \end{aligned} \quad (3.57)$$

with:

$$C \triangleq \mathbf{1}_{\tilde{n}} \otimes I_{d_z}, \quad \mathbf{1}_{\tilde{n}} \triangleq \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^{\tilde{n}} \quad (3.58)$$

The ordinary kriging weights thus minimise (3.42) under the constraint (3.57). Using the Lagrange multiplier method, Proposition 3.8 derives the corresponding estimator and covariance matrix, and Remark 3.8 establishes the link between ordinary and simple kriging.

Proposition 3.8 (Ordinary kriging estimator). *The ordinary kriging estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_o(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) + k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} (\tilde{\mathbf{z}} - C m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ \hat{R}_o(x, \tilde{\mathbf{x}}) = \hat{R}_{m_o}(x, \tilde{\mathbf{x}}) + k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \quad (3.59)$$

with:

$$\begin{cases} m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = (C^\top G(\tilde{\mathbf{x}})^{-1} C)^{-1} C^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ \hat{R}_{m_o}(x, \tilde{\mathbf{x}}) = (I_{d_z} - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} C) (C^\top G(\tilde{\mathbf{x}})^{-1} C)^{-1} \\ \quad (I_{d_z} - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} C)^\top \end{cases} \quad (3.60)$$

Proof. Let:

$$\mathcal{L}(W, \xi) \triangleq \text{tr} (W^\top \tilde{\mathbf{z}} - z) + 2 \text{tr} (\xi^\top (C^\top W - I_{d_z})) \quad (3.61)$$

be the Lagrangian associated with the optimisation problem (3.42) under the constraint (3.57), and $\xi \in \mathcal{M}_{d_z, d_z}(\mathbb{R})$ the corresponding Lagrange multiplier.

This problem is solved by finding the minimum of $\mathcal{L}$, *i.e.* finding a zero of its derivative. Writing G for $G(\tilde{\mathbf{x}})$, the same properties used in the proof of Proposition 3.5 give:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial W}(W, \xi) &= 0 \\ \iff 2G W - 2k(\tilde{\mathbf{x}}, x) + 2 \frac{\partial}{\partial W} (\text{tr} (\xi^\top (C^\top W - I_{d_z}))) &= 0 \\ \iff 2G W - 2k(\tilde{\mathbf{x}}, x) + 2C\xi &= 0 \\ \iff G W + C\xi &= k(\tilde{\mathbf{x}}, x) \end{aligned} \quad (3.62)$$

Together with the constraint $C^\top W = I_{d_z}$, this yields the block system:

$$\begin{bmatrix} G & C \\ C^\top & 0 \end{bmatrix} \begin{bmatrix} W \\ \xi \end{bmatrix} = \begin{bmatrix} k(\tilde{\mathbf{x}}, x) \\ I_{d_z} \end{bmatrix} \quad (3.63)$$

The left matrix block can be inverted using the upper left Schur inversion formula from Proposition A.2. Its Schur complement,

$$S \triangleq 0 - C^\top G^{-1} C = -C^\top G^{-1} C, \quad (3.64)$$

is invertible since $G \succ 0$ and C has full column rank. The convention of Proposition A.2 gives the top block row of the inverse:

$$\bar{A} = G^{-1} + G^{-1} C S^{-1} C^\top G^{-1}, \quad \bar{B} = -G^{-1} C S^{-1} \quad (3.65)$$

In particular, it comes that:

$$\begin{aligned} W &= \bar{A} k(\tilde{\mathbf{x}}, x) + \bar{B} I_{d_z} \\ &= (G^{-1} + G^{-1} C S^{-1} C^\top G^{-1}) k(\tilde{\mathbf{x}}, x) - G^{-1} C S^{-1} I_{d_z} \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C S^{-1} (C^\top G^{-1} k(\tilde{\mathbf{x}}, x) - I_{d_z}) \\ &= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C (C^\top G^{-1} C)^{-1} (I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)) \end{aligned} \quad (3.66)$$

The ordinary kriging estimator is thus:

$$\begin{aligned}\hat{z}_o &= W^\top \tilde{\mathbf{z}} \\ &= k(x, \tilde{\mathbf{x}}) G^{-1} \tilde{\mathbf{z}} + (I_{d_z} - k(x, \tilde{\mathbf{x}}) G^{-1} C) (C^\top G^{-1} C)^{-1} C^\top G^{-1} \tilde{\mathbf{z}} \\ &= (C^\top G^{-1} C)^{-1} C^\top G^{-1} \tilde{\mathbf{z}} + k(x, \tilde{\mathbf{x}}) G^{-1} (\tilde{\mathbf{z}} - C (C^\top G^{-1} C)^{-1} C^\top G^{-1} \tilde{\mathbf{z}})\end{aligned}\quad (3.67)$$

The covariance matrix is obtained in a similar fashion. **TODO: à rajouter quand-même ?** $\square$

Remark 3.8 (Comparison between simple and ordinary kriging). *The formulas obtained in (3.59) closely resemble those of the simple kriging. In fact, they can be rewritten as follows:*

$$\begin{cases} \hat{z}_o(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{z}_o(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ \hat{R}_o(x, \tilde{\mathbf{x}}) = \hat{R}_{m_o}(x, \tilde{\mathbf{x}}) + \hat{R}_s(x, \tilde{\mathbf{x}}) \end{cases} \quad (3.68)$$

where, by abuse of notation, $m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ denotes either the estimated mean vector or its replication for each the training point $\tilde{\mathbf{x}}$, namely:

$$\underbrace{[m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}), \dots, m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})]}_{\tilde{n} \text{ times}} = C m_o(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \quad (3.69)$$

Ordinary kriging can therefore be interpreted as simple kriging in which the mean m_o is concurrently estimated from the observations. The uncertainty associated with this second estimation is captured by the covariance term $\hat{R}_{m_o} \succeq 0$, and is added to the simple kriging covariance.

3.3.3 Universal Kriging

TODO: on en a besoin pour 3.5.1 mais jsp si il faut démontrer tous les résultats [69]

3.4 Reduced-Rank Kernel Approximation

Exact GP regression requires inverting the $\tilde{n}d_z \times \tilde{n}d_z$ Gram matrix $G(\tilde{\mathbf{x}})$, which scales as $\mathcal{O}(\tilde{n}^3)$. When the number of training points is large, this operation becomes prohibitive. Reduced-rank methods [70, 71] circumvent this issue by approximating the kernel as a linear combination of $r \in \mathbb{N}^*$ basis functions, with $r < \tilde{n}$. The inversion cost is reduced to $\mathcal{O}(r^3)$, bringing down the overall kriging cost to $\mathcal{O}(r^2 \tilde{n})$.

Mercer's theorem [72], presented in Proposition 3.10, establishes the existence of such a decomposition as soon as k is a continuous SPD kernel, which Matérn kernels satisfy, as proven by Proposition 3.3, and $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. The Woodbury inversion formula is then used to rewrite the kriging estimator in a computationally efficient expression.

3.4.1 Mercer's Theorem

We start by generalising Definition 3.2 to functional operators on $\mathcal{L}^2(\mathcal{U})$.

Definition 3.5 (Operator self-adjoint positive semi-definiteness). *A functional operator T on $\mathcal{L}^2(\mathcal{U})$ is self-adjoint positive semi-definite (SPSD) if:*

$$\begin{cases} \forall f, g \in \mathcal{L}^2(\mathcal{U}), \langle Tf|g \rangle = \langle f|Tg \rangle & \text{self-adjoint} \\ \forall f \in \mathcal{L}^2(\mathcal{U}), \langle f|Tf \rangle \geq 0 & \text{positive semi-definite} \end{cases} \quad (3.70)$$

where $\langle \cdot | \cdot \rangle$ is the following scalar product:

$$\forall f, g \in \mathcal{L}^2(\mathcal{U}) \mapsto \langle f|g \rangle \triangleq \int_{\mathcal{U}} f(x)^\top g(x) dx \quad (3.71)$$

Proposition 3.9 (Mercer operator). *Assume that $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$ and k is a continuous SPD kernel. The Mercer operator:*

$$\mathcal{K} : \begin{cases} \mathcal{L}^2(\mathcal{U}) \rightarrow \mathcal{L}^2(\mathcal{U}) \\ f \mapsto \left(x \mapsto \int_{\mathcal{U}} k(x, y) f(y) dy \right) \end{cases} \quad (3.72)$$

is a compact SPD functional operator.

Proof. As $\mathcal{U}$ is compact and k continuous, k is bounded on $\mathcal{U} \times \mathcal{U}$, and $\mathcal{K}$ is compact.

Let $f, g \in \mathcal{L}^2(\mathcal{U})$. Cauchy-Schwarz inequality ensures that there exists a constant $c \in \mathbb{R}_+$ such that:

$$\iint_{\mathcal{U} \times \mathcal{U}} |f(y)^\top k(x, y)^\top g(x)| dx dy \leq c \sqrt{\langle f|f \rangle} \sqrt{\langle g|g \rangle} < +\infty \quad (3.73)$$

Fubini's integration theorem can thus be used to give:

$$\begin{aligned} \langle \mathcal{K}f|g \rangle &= \int_{\mathcal{U}} (\mathcal{K}f)(x)^\top g(x) dx \\ &= \int_{\mathcal{U}} \left(\int_{\mathcal{U}} k(x, y) f(y) dy \right)^\top g(x) dx \\ &= \int_{\mathcal{U}} \int_{\mathcal{U}} f(y)^\top k(x, y)^\top g(x) dy dx \\ &= \int_{\mathcal{U}} \int_{\mathcal{U}} f(y)^\top k(y, x) g(x) dx dy \\ &= \int_{\mathcal{U}} f(y)^\top \left(\int_{\mathcal{U}} k(y, x) g(x) dx \right) dy \\ &= \int_{\mathcal{U}} f(y)^\top (\mathcal{K}g)(y) dy \\ &= \langle f|\mathcal{K}g \rangle \end{aligned} \quad (3.74)$$

Let $f : \mathcal{U} \rightarrow \mathbb{R}^{d_z}$ be a continuous function, and let $n \in \mathbb{N}^*$. We define $(\mathcal{U}_i^{(n)})_{i \in \llbracket 1, n \rrbracket}$ a partition of $\mathcal{U}$ such that the diameter of each $\mathcal{U}_i^{(n)}$, $i \in \llbracket 1, n \rrbracket$ tends to 0 when $n \rightarrow +\infty$. For $i \in \llbracket 1, n \rrbracket$, we define $x_i^{(n)} \in \mathcal{U}_i^{(n)}$, and we let:

$$a_i^{(n)} \triangleq \left(\int_{\mathcal{U}_i^{(n)}} dx \right) f(x_i^{(n)}) \quad (3.75)$$

By continuity of f , the function:

$$(x, y) \mapsto f(x)^\top k(x, y) f(y) \quad (3.76)$$

is continuous on the compact $\mathcal{U} \times \mathcal{U}$, so its Riemann sums converge:

$$\sum_{i=1}^n \sum_{j=1}^n \left(a_i^{(n)} \right)^\top k \left(x_i^{(n)}, x_j^{(n)} \right) a_j^{(n)} \xrightarrow{n \rightarrow +\infty} \iint_{\mathcal{U} \times \mathcal{U}} f(x)^\top k(x, y) f(y) dx dy = \langle f | \mathcal{K} f \rangle \quad (3.77)$$

The SPSPDness of k ensures that each Riemann sum is non-negative. By passing to the limit, it follows that:

$$\langle f | \mathcal{K} f \rangle \geq 0 \quad (3.78)$$

for every continuous function f .

Finally, recall that the set of continuous functions of $\mathcal{U}$ is dense in $\mathcal{L}^2(\mathcal{U})$. By continuity of:

$$f \in \mathcal{L}^2(\mathcal{U}) \mapsto \langle f | \mathcal{K} f \rangle \quad (3.79)$$

the inequality (3.78) extends to all $f \in \mathcal{L}^2(\mathcal{U})$. $\square$

Proposition 3.10 (Mercer's theorem). *Under the assumptions of Proposition 3.9, there exist a non-increasing sequence $(\lambda_i)_{i \in \mathbb{N}^*}$ of strictly positive eigenvalues of $\mathcal{K}$ converging to 0, and an associated orthonormal family $(\psi_i)_{i \in \mathbb{N}^*}$ of continuous $\mathbb{R}^{d_z}$ -valued eigenfunctions, such that:*

$$\forall x, x' \in \mathcal{U}, k(x, x') = \sum_{i=1}^{+\infty} \lambda_i \psi_i(x) \psi_i(x')^\top \quad (3.80)$$

Furthermore, this series converges absolutely and uniformly on $\mathcal{U} \times \mathcal{U}$.

Proof. By Proposition 3.9, $\mathcal{K}$ is a compact and self-adjoint operator on $\mathcal{L}^2(\mathcal{U})$. The spectral theorem [73] thus states that there exists an orthonormal basis $(\psi_i)_{i \in \mathbb{N}^*}$ of $\mathcal{L}^2(\mathcal{U})$ consisting of eigenfunctions of $\mathcal{K}$, with each corresponding eigenvalue $(\lambda_i)_{i \in \mathbb{N}^*}$ being real. Up to reordering, the sequence of eigenvalues can be considered to be non-increasing.

Since k is uniformly continuous on the compact $\mathcal{U} \times \mathcal{U}$, $\mathcal{K}f$ is continuous for every $f \in \mathcal{L}^2(\mathcal{U})$. In particular, for each $i \in \mathbb{N}^*$ such that $\lambda_i \neq 0$, the eigenfunction:

$$\psi_i = \lambda_i^{-1} \mathcal{K} \psi_i \quad (3.81)$$

is continuous. Those associated with a null eigenvalue do not contribute to (3.80) and may be disregarded.

Since $(\psi_i)_{i \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{L}^2(\mathcal{U})$, the family $(\psi_i \psi_j^\top)_{i, j \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{L}^2(\mathcal{U} \times \mathcal{U})$. Let $\langle \cdot | \cdot \rangle_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}$ denote:

$$\forall f, g \in \mathcal{L}^2(\mathcal{U} \times \mathcal{U}) \mapsto \langle f | g \rangle_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})} \triangleq \iint_{\mathcal{U} \times \mathcal{U}} \text{tr} (f(x, y)^\top g(x, y)) dx dy \quad (3.82)$$

Let $i, j \in \mathbb{N}^*$. We have:

$$\begin{aligned}
\langle k | \psi_i \psi_j^\top \rangle_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})} &= \iint_{\mathcal{U} \times \mathcal{U}} \text{tr} (k(x, y)^\top \psi_i(x) \psi_j(y)^\top) dx dy \\
&= \iint_{\mathcal{U} \times \mathcal{U}} \psi_j(y)^\top k(x, y)^\top \psi_i(x) dx dy \\
&= \int_{\mathcal{U}} \psi_j(y)^\top \left(\int_{\mathcal{U}} k(y, x) \psi_i(x) dx \right) dy \\
&= \int_{\mathcal{U}} \psi_j(y)^\top \mathcal{K} \psi_i(y) dy \\
&= \lambda_i \int_{\mathcal{U}} \psi_j(y)^\top \psi_i(y) dy \\
&= \lambda_i \delta_{i,j}
\end{aligned} \tag{3.83}$$

where:

$$\delta_{i,j} \triangleq \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{3.84}$$

is the Kronecker symbol.

Since $k \in \mathcal{L}^2(\mathcal{U} \times \mathcal{U})$, it follows from the completeness of $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ that:

$$k_r(x, x') \triangleq \sum_{i=1}^r \lambda_i \psi_i(x) \psi_i(x')^\top \tag{3.85}$$

converges to k in the $\mathcal{L}^2(\mathcal{U} \times \mathcal{U})$ sense.

It must be noted that $\mathcal{L}^2$ convergence only identifies k up to a Lebesgue-null subset of $\mathcal{U} \times \mathcal{U}$, and gives no control at any specific point of the domain. In particular, it is not enough to justify substituting k_r for k in kriging scenarios. However, since k is continuous on $\mathcal{U} \times \mathcal{U}$, the convergence can be extended to absolute and uniform convergence on the entire $\mathcal{U} \times \mathcal{U}$. While we will not dive into the details of this demonstration, it is obtained via Dini's theorem applied to the non-decreasing sequence $(x \mapsto \text{tr } k_r(x, x))_{r \in \mathbb{N}^*}$.

Since $\mathcal{K}$ is SPSPD, every eigenvalue is non-negative. Assume that there exists a rank $r_0 \in \mathbb{N}^*$ such that $\lambda_i = 0$ as soon as $i > r_0$. Then the decomposition (3.80) would reduce to the finite sum:

$$k = \sum_{i=1}^{r_0} \lambda_i \psi_i \psi_i^\top \tag{3.86}$$

whose Gram matrices have rank at most r_0 and are thus singular at any n distinct points with $nd_z > r_0$, contradicting the SPDness of k by Proposition 3.2. Consequently:

$$\lambda_1 \geq \lambda_2 \geq \dots > 0 \tag{3.87}$$

□

Remark 3.9 (Definiteness without invertibility). *Since all the eigenvalues λ_i are strictly positive, the quadratic form:*

$$f \in \mathcal{L}^2(\mathcal{U}) \mapsto \langle f | \mathcal{K} f \rangle = \sum_{i=1}^{+\infty} \lambda_i \langle f | \psi_i \rangle^2 \tag{3.88}$$

is strictly positive for every $f \neq 0$, hence positive definite. For a symmetric matrix, definiteness amounts to invertibility, but this equivalence fails in infinite dimension. In particular, since $\mathcal{K}$ is compact with $\lambda_i \rightarrow 0$, it admits no positive lower bound on the unit sphere, so 0 belongs to its spectrum. $\mathcal{K}^{-1}$ is thus unbounded, and so the Mercer operator is positive definite yet not invertible.

In finite dimension, strict positivity and invertibility coincide and jointly characterise SPD matrices, whereas this is not the case in infinite dimension. This is not restrictive, however, as only the finite Gram matrices induced by k are inverted throughout this document.

3.4.2 Efficient Kriging Estimation

Proposition 3.11 (Reduced-rank approximation). *For a continuous SPD kernel k , we define the rank of k as the dimension of the image of $\mathcal{L}^2(\mathcal{U})$ by its associated Mercer operator:*

$$\text{rank}(k) \triangleq \dim(\mathcal{K}(\mathcal{L}^2(\mathcal{U}))) \quad (3.89)$$

Under the assumptions of Proposition 3.9, the truncated kernel:

$$k_r : \begin{cases} \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{M}_{d_z, d_z}(\mathbb{R}) \\ x, x' \mapsto \sum_{i=1}^r \lambda_i \psi_i(x) \psi_i(x')^\top \end{cases} \quad (3.90)$$

is a continuous SPSPD kernel of rank at most $r \in \mathbb{N}^*$. The sequence $(k_r)_{r \in \mathbb{N}^*}$ converges uniformly towards k as $r \rightarrow +\infty$, and k_r minimises $\|k - k_r\|_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}$ among all kernels of rank at most r .

Proof. Let $r \in \mathbb{N}^*$ and $f \in \mathcal{L}^2(\mathcal{U})$. We have:

$$\mathcal{K}_r f = \sum_{i=1}^r \lambda_i \langle f | \psi_i \rangle \psi_i \quad (3.91)$$

where $\mathcal{K}_r$ is the Mercer operator associated with k_r . Every $\mathcal{K}_r f$ thus belongs to the subspace of continuous functions spanned by $\psi_1, \dots, \psi_r$. The orthonormality of the basis ensures that this subspace is of dimension r , and thus the rank of k_r is at most r . The uniform convergence of $(k_r)_{r \in \mathbb{N}^*}$ has already been proven by Proposition 3.10.

Finally, applying the Eckart-Young-Mirsky theorem [74, 75] to $\mathcal{K}$ and using the orthonormality of $(\psi_i \psi_j^\top)_{i,j \in \mathbb{N}^*}$ yields:

$$\min_{\substack{\tilde{k} \in \mathcal{L}^2(\mathcal{U} \times \mathcal{U}) \\ \text{rank}(\tilde{k}) \leq r}} \|k - \tilde{k}\|_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}^2 = \sum_{j>r} \lambda_j^2 = \|k - k_r\|_{\mathcal{L}^2(\mathcal{U} \times \mathcal{U})}^2 \quad (3.92)$$

□

In compact form, the eigenfunctions are collected into:

$$\Psi(x) \triangleq [\psi_1(x) \quad \cdots \quad \psi_r(x)] \in \mathcal{M}_{d_z, r}(\mathbb{R}) \quad (3.93)$$

and the corresponding eigenvalues are gathered as:

$$\Lambda \triangleq \begin{bmatrix} \lambda_1 & & (0) \\ & \ddots & \\ (0) & & \lambda_r \end{bmatrix} \in \mathcal{M}_{r, r}(\mathbb{R}) \quad (3.94)$$

By Proposition 3.10, the eigenvalues λ_i are strictly positive, so that $\Lambda \succ 0$ and Λ^{-1} is well-defined. Proposition 3.11 can thus be rewritten as:

$$k(x, x') \approx \Psi(x) \Lambda \Psi(x')^\top \quad (3.95)$$

and then:

$$G(\tilde{\mathbf{x}}) \approx \Psi(\tilde{\mathbf{x}}) \Lambda \Psi(\tilde{\mathbf{x}})^\top + I_{\tilde{n}} \otimes \tilde{R}, \quad \Psi(\tilde{\mathbf{x}}) \triangleq \begin{bmatrix} \Psi(\tilde{x}_1) \\ \vdots \\ \Psi(\tilde{x}_{\tilde{n}}) \end{bmatrix} \quad (3.96)$$

Proposition 3.12 (Reduced-rank simple kriging). *Let:*

$$\tilde{G}(\tilde{\mathbf{x}}) \triangleq \Lambda^{-1} + \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{n}} \otimes \tilde{R} \right)^{-1} \Psi(\tilde{\mathbf{x}}) \in \mathcal{M}_{r,r}(\mathbb{R}) \quad (3.97)$$

be the reduced-rank Gram matrix. Under the reduced-rank approximation of (3.96), the centred simple kriging estimator of Proposition 3.5 and its error covariance matrix reduce to:

$$\begin{cases} \hat{z}_s(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{n}} \otimes \tilde{R} \right)^{-1} \tilde{\mathbf{z}} \\ \hat{R}_s(x, \tilde{\mathbf{x}}) \approx R + \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{cases} \quad (3.98)$$

which require only the inversion of $\tilde{G}(\tilde{\mathbf{x}})$.

When there exists $\sigma > 0$ such that $\tilde{R} = \sigma^2 I$, (3.98) further simplify to:

$$\begin{cases} \hat{z}_s(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \Psi(\tilde{\mathbf{x}})^\top \tilde{\mathbf{z}} \\ \hat{R}_s(x, \tilde{\mathbf{x}}) \approx R + \sigma^2 \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \Psi(x)^\top \end{cases} \quad (3.99)$$

These expressions remain valid at $\sigma = 0$ provided $\Psi(\tilde{\mathbf{x}})$ has full column rank.

Proof. As (3.96) is a rank r perturbation of $A \triangleq I_{\tilde{n}} \otimes \tilde{R}$, the Woodbury inversion formula (see Proposition A.5) can be used to reduce the $\tilde{n}d_z \times \tilde{n}d_z$ inversion of $G(\tilde{\mathbf{x}})$ to an $r \times r$ one:

$$G(\tilde{\mathbf{x}})^{-1} \approx A^{-1} - A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \quad (3.100)$$

Computing the centred simple kriging estimator from Proposition 3.5 using this approximation and the push-through identity (see Proposition A.4) yields:

$$\begin{aligned} \hat{z}_s(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) &= k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ &\approx \Psi(x) \Lambda \Psi(\tilde{\mathbf{x}})^\top \left(I - A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \right) A^{-1} \tilde{\mathbf{z}} \\ &= \Psi(x) \Lambda \left(I - \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} \right) \Psi(\tilde{\mathbf{x}})^\top A^{-1} \tilde{\mathbf{z}} \end{aligned} \quad (3.101)$$

Recall that, by definition of $\tilde{G}(\tilde{\mathbf{x}})$:

$$\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} = \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \quad (3.102)$$

which gives:

$$\begin{aligned} I - \Psi(\tilde{\mathbf{x}})^\top A^{-1} \Psi(\tilde{\mathbf{x}}) \tilde{G}(\tilde{\mathbf{x}})^{-1} &= I - \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} \right) \tilde{G}(\tilde{\mathbf{x}})^{-1} \\ &= I - I + \Lambda^{-1} \tilde{G}(\tilde{\mathbf{x}})^{-1} \\ &= \Lambda^{-1} \tilde{G}(\tilde{\mathbf{x}})^{-1} \end{aligned} \quad (3.103)$$

Plugging this result into (3.101) and using $A^{-1} = I_{\tilde{n}} \otimes \tilde{R}^{-1}$ finally gives:

$$\hat{z}_s(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \tilde{\mathbf{z}} \quad (3.104)$$

For $\hat{R}_s$, reusing:

$$k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} \approx \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \quad (3.105)$$

together with (3.103) yields:

$$\begin{aligned} \hat{R}_s &= k(x, x) + R - k(x, \tilde{\mathbf{x}}) G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &\approx \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top A^{-1} \times \Psi(\tilde{\mathbf{x}}) \Lambda \Psi(x)^\top \\ &= \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} \right) \Lambda \Psi(x)^\top \\ &= \Psi(x) \Lambda \Psi(x)^\top + R - \Psi(x) \left(\Lambda - \tilde{G}(\tilde{\mathbf{x}})^{-1} \right) \Psi(x)^\top \\ &= R + \Psi(x) \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{aligned} \quad (3.106)$$

Finally, when $\tilde{R} = \sigma^2 I_{d_z}$ with $\sigma > 0$, substituting $A^{-1} = \sigma^{-2} I_{nd_z}$ yields the announced simplifications. For such $\sigma > 0$, the matrix:

$$\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \succ 0 \quad (3.107)$$

is SPD, as the sum of the SPSD matrix $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}})$ and the SPD matrix $\sigma^2 \Lambda^{-1}$. The map:

$$\sigma \in \mathbb{R}_+^* \mapsto \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \right)^{-1} \quad (3.108)$$

is thus well-defined and continuous, and extends to $\sigma = 0$ when $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) \succ 0$, *i.e.* when $\Psi(\tilde{\mathbf{x}})$ has full column rank. As $\hat{z}_s$ and $\hat{R}_s$ depend on σ only through this inverse and an explicit factor σ^2 , both expressions remain valid at $\sigma = 0$ under this condition. $\square$

Remark 3.10 (Construction of the eigenfunctions). *While Mercer's theorem establishes the existence of the decomposition presented by (3.80), it provides no means to compute a closed-form expression of k_r . In practice [76], additional assumptions on k , namely radial properties, are usually made in order to derive an explicit approximation of the form of Proposition 3.11.*

3.5 Kernel-Induced Functional Representations

Beyond regression, the covariance kernel induces functional representations exploited later in this thesis. Section 3.4 replaces the GP by a finite-dimensional projection to alleviate the cubic cost of kriging, and section 3.5.3 interprets kriging as minimum-norm interpolation in a reproducing kernel Hilbert space.

3.5.1 Universal Kriging Mean

TODO + kriging update formulas

3.5.2 Karhunen-Loeve Decomposition

TODO [77]

3.5.3 Reproducing Kernel Hilbert Space Formulation

The covariance kernel also induces a deterministic, function-analytic counterpart to GP regression. Let $k : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}$ be a scalar SPSPD kernel on an arbitrary index set $\mathcal{U}$ [78, 79].

Definition 3.6 (Reproducing kernel Hilbert space). *The reproducing kernel Hilbert space (RKHS) $\mathcal{H}_k$ associated with k is the completion of the span of $\{k(\cdot, x) \mid x \in \mathcal{U}\}$ for the inner product defined on the generators by:*

$$\langle k(\cdot, x) \mid k(\cdot, x') \rangle_{\mathcal{H}_k} \triangleq k(x, x') \quad (3.109)$$

It is characterised by the reproducing property. For all $\psi \in \mathcal{H}_k$ and $x \in \mathcal{U}$,

$$\psi(x) = \langle \psi \mid k(\cdot, x) \rangle_{\mathcal{H}_k} \quad (3.110)$$

The reproducing property (3.110) states that point evaluation is a continuous linear functional on $\mathcal{H}_k$, which endows the space with a regularity entirely dictated by k . This space provides a variational reading of kriging [80].

Proposition 3.13 (Kriging as minimum-norm interpolation). *In the centred case, the simple kriging estimator (3.44) is the solution of the penalised least-squares problem in $\mathcal{H}_k$:*

$$z_s^* = \underset{\psi \in \mathcal{H}_k}{\operatorname{argmin}} \left(\frac{1}{\tilde{R}} \sum_{i=1}^{\tilde{n}} (\psi(\tilde{x}_i) - \tilde{z}_i)^2 + \|\psi\|_{\mathcal{H}_k}^2 \right) \quad (3.111)$$

which, in the noise-free limit $\tilde{R} \rightarrow 0$, reduces to the minimum-norm interpolant $\operatorname{argmin}\{\|\psi\|_{\mathcal{H}_k} \mid \psi(\tilde{x}_i) = \tilde{z}_i\}$.

Proof. By the representer theorem [81, 82], a minimiser of (3.111) lies in the finite-dimensional span of $\{k(\cdot, \tilde{x}_i)\}_{i=1}^{\tilde{n}}$, i.e. $\psi = \sum_i \alpha_i k(\cdot, \tilde{x}_i)$. The reproducing property gives $\psi(\tilde{x}_j) = \sum_i \alpha_i k(\tilde{x}_j, \tilde{x}_i)$ and $\|\psi\|_{\mathcal{H}_k}^2 = \alpha^\top k(\tilde{X}, \tilde{X}) \alpha$, so (3.111) becomes the quadratic problem $\operatorname{argmin}_\alpha \tilde{R}^{-1} \|k(\tilde{X}, \tilde{X}) \alpha - \tilde{Z}\|^2 + \alpha^\top k(\tilde{X}, \tilde{X}) \alpha$. Cancelling its gradient gives $\alpha = G^{-1} \tilde{Z}$, hence $\psi(x) = k(x, \tilde{X}) G^{-1} \tilde{Z}$, which is (3.44) for $m = 0$. $\square$

Remark 3.11 (GP and RKHS correspondence). *The two viewpoints are complementary [83]. The GP model gives a probabilistic reading of kriging (posterior mean and variance under a Gaussian prior), while the RKHS gives a deterministic one (minimum-norm interpolant in a function space). Writing the Mercer decomposition of the kernel over its eigenpairs $(\beta_j, \phi_j)_{j \geq 1}$ as $k(x, x') = \sum_{j \geq 1} \beta_j \phi_j(x) \phi_j(x')$, a function $\psi = \sum_{j \geq 1} c_j \phi_j$ belongs to $\mathcal{H}_k$ with squared norm $\|\psi\|_{\mathcal{H}_k}^2 = \sum_{j \geq 1} c_j^2 / \beta_j$; the norm thus penalises most the components carried by low-eigenvalue, rapidly oscillating eigenfunctions, so the decay of the eigenvalues β_j sets the smoothness enforced by k . Choosing a kernel therefore amounts to choosing a regularity class for the unknown field. A smooth kernel such as the SE kernel yields infinitely differentiable interpolants, whereas a rougher Matérn kernel admits functions of limited smoothness.*

Espaces natifs : [84]

Part II

Navigating on Gaussian Processes

CHAPTER 4

THE CLUSTERED ADAPTIVE KRIGING PARTICLE
 FILTER

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4.1 Introduction

[85, 86, 87]

4.2 Ablation Study

4.2.1 Influence of the Training Data

cf poster + FUSION 2024

4.2.2 Influence of the Subsampling Window

subsampling methods + learning the hyperparameters

4.2.3 Influence of the Clustering Step

clustering methods + EUSIPCO 2025

4.3 Theoretical Subsampling Error

TODO: on essaie tjs de publier l'article ?

Part III

Appendix

APPENDIX A

USEFUL MATRIX IDENTITIES

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A.1 Gaussian Conditioning Formula

Proposition A.1 (Gaussian conditioning [88]). *For $n, p \in \mathbb{N}$, let $U \in \mathbb{R}^n$ and $V \in \mathbb{R}^p$ be jointly Gaussian random variables, of respective means μ_U, μ_V and covariance matrices $A \succ 0$ and $D \succ 0$. There exists $B \in \mathcal{M}_{n,p}(\mathbb{R})$ such that:*

$$\begin{bmatrix} U \\ V \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mu_U \\ \mu_V \end{bmatrix}, \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \right) \quad (\text{A.1})$$

We assume B to be such that this joint covariance is SPD in the sense of Definition 2.1. The conditional distribution of U given $V = v$ is then:

$$U|V = v \sim \mathcal{N}(\mu_U + BD^{-1}(v - \mu_V), P_{U|V}) \quad (\text{A.2})$$

with:

$$P_{U|V} \triangleq A - BD^{-1}B^\top \succ 0 \quad (\text{A.3})$$

Proof. The existence of B is direct by definition of jointly Gaussian random variables. As $D \succ 0$ is invertible, $S \triangleq A - BD^{-1}B^\top$ the Schur complement with respect to D is well-defined.

Let $u \neq 0$:

$$\begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix}^\top \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix} = u^\top (A - BD^{-1}B^\top) u > 0 \quad (\text{A.4})$$

by SPDness of the joint covariance, the left-hand vector being non-zero. S is thus also invertible, and we can apply the inferior right Schur inversion formula (see Proposition A.2) to obtain:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \triangleq \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}^{-1} \quad (\text{A.5})$$

For some realisation u of U and v of V , we define $\tilde{u} \triangleq u - \mu_U$ and $\tilde{v} \triangleq v - \mu_V$, and we compute:

$$\begin{aligned} \alpha &\triangleq \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix}^\top \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix} \\ &= \tilde{u}^\top \bar{A} \tilde{u} + \tilde{u}^\top \bar{B} \tilde{v} + \tilde{v}^\top \bar{C} \tilde{u} + \tilde{v}^\top \bar{D} \tilde{v} \\ &= \tilde{u}^\top S^{-1} \tilde{u} + \tilde{u}^\top S^{-1} (-BD^{-1} \tilde{v}) + (-BD^{-1} \tilde{v})^\top S^{-1} \tilde{u} + \\ &\quad (-BD^{-1} \tilde{v})^\top S^{-1} (-BD^{-1} \tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \\ &= (\tilde{u} - BD^{-1} \tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1} \tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \end{aligned} \quad (\text{A.6})$$

The conditional density is thus:

$$\begin{aligned} p(\tilde{u}|\tilde{v}) &\propto \frac{p(\tilde{u}, \tilde{v})}{p(\tilde{v})} \\ &\propto \frac{\exp\left(-\frac{1}{2}\alpha\right)}{\exp\left(-\frac{1}{2}\tilde{v}^\top D^{-1} \tilde{v}\right)} \\ &\propto \exp\left(-\frac{1}{2}(\tilde{u} - BD^{-1} \tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1} \tilde{v})\right) \end{aligned} \quad (\text{A.7})$$

With this formulation, it comes that $U|V = v$ is Gaussian with mean:

$$\mu_{U|V} \triangleq \mu_U + BD^{-1}(v - \mu_V) \quad (\text{A.8})$$

and covariance:

$$P_{U|V} \triangleq S = A - BD^{-1}B^\top \succ 0 \quad (\text{A.9})$$

□

A.2 Schur Inversion Formula

Proposition A.2 (Schur inversion formula [89]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$, with A invertible. We define the matrix M and its Schur complement with respect to A as:*

$$M \triangleq \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \quad M/A \triangleq D - CA^{-1}B \quad (\text{A.10})$$

If M/A is invertible, then M is invertible and:

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} & -A^{-1}B(M/A)^{-1} \\ -(M/A)^{-1}CA^{-1} & (M/A)^{-1} \end{bmatrix} \quad (\text{A.11})$$

Proof. Let $\bar{M}$ be the matrix defined by the right-hand side of the formula, and write its blocks as:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} = \bar{M} \quad (\text{A.12})$$

The identity $M\bar{M} = I$ yields:

$$\begin{cases} A\bar{A} + B\bar{C} = I \\ A\bar{B} + B\bar{D} = 0 \\ C\bar{A} + D\bar{C} = 0 \\ C\bar{B} + D\bar{D} = I \end{cases} \quad (\text{A.13})$$

In particular, the second equation can be rewritten as:

$$\bar{B} = -A^{-1}B\bar{D} \quad (\text{A.14})$$

which can be plugged into the fourth equation to give:

$$\begin{aligned} & C\bar{B} + D\bar{D} = I \\ \iff & -CA^{-1}B\bar{D} + D\bar{D} = I \\ \iff & (D - CA^{-1}B)\bar{D} = I \\ \iff & (M/A)\bar{D} = I \\ \iff & \bar{D} = (M/A)^{-1} \end{aligned} \quad (\text{A.15})$$

(A.14) can then be solved:

$$\bar{B} = -A^{-1}B(M/A)^{-1} \quad (\text{A.16})$$

A similar operation can be performed on the two remaining equations to obtain:

$$\bar{A} = A^{-1}(I - B\bar{C}) \quad (\text{A.17})$$

and:

$$\begin{aligned} & C\bar{A} + D\bar{C} = 0 \\ \iff & CA^{-1}(I - B\bar{C}) + D\bar{C} = 0 \\ \iff & CA^{-1} + (D - CA^{-1}B)\bar{C} = 0 \\ \iff & CA^{-1} + (M/A)\bar{C} = 0 \\ \iff & \bar{C} = -(M/A)^{-1}CA^{-1} \end{aligned} \quad (\text{A.18})$$

Finally:

$$\bar{A} = A^{-1}(I - B\bar{C}) = A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} \quad (\text{A.19})$$

It can be verified by direct calculations that:

$$M\bar{M} = \bar{M}M = I \quad (\text{A.20})$$

which proves that M is indeed invertible with inverse $\bar{M}$. $\square$

Remark A.1 (Inversion with respect to D). *If D is invertible, one can instead introduce the Schur complement with respect to D :*

$$M/D \triangleq A - BD^{-1}C \quad (\text{A.21})$$

When M/D is also invertible, an analogous block inversion formula holds:

$$M^{-1} = \begin{bmatrix} (M/D)^{-1} & -(M/D)^{-1}BD^{-1} \\ -D^{-1}C(M/D)^{-1} & D^{-1} + D^{-1}C(M/D)^{-1}BD^{-1} \end{bmatrix} \quad (\text{A.22})$$

Remark A.2 (Combining both formulations). *When both A and D are invertible, the Woodbury lemma proves that the two formulations give the same matrices for each block. In particular, they can be combined block by block at will.*

A.3 Woodbury Inversion Formula

Proposition A.3 (Inversion identity). *For $n \in \mathbb{N}$, let $P \in \mathcal{M}_{n,n}(\mathbb{R})$ such that $I + P$ is invertible. Then:*

$$(I + P)^{-1} = I - (I + P)^{-1}P \quad (\text{A.23})$$

Proof. This result is direct from:

$$I = (I + P)^{-1}(I + P) = (I + P)^{-1} + (I + P)^{-1}P \quad (\text{A.24})$$

$\square$

Proposition A.4 (Push-through identity [90]). *For $n, p \in \mathbb{N}$, let $P \in \mathcal{M}_{n,p}(\mathbb{R})$ and $Q \in \mathcal{M}_{p,n}(\mathbb{R})$ such that $I_n + PQ$ is invertible. Then $I_p + QP$ is also invertible and:*

$$(I_n + PQ)^{-1}P = P(I_p + QP)^{-1} \quad (\text{A.25})$$

Proof. We start by using the identity:

$$P + PQP = P(I_p + QP) = (I_n + PQ)P \quad (\text{A.26})$$

Thus:

$$\begin{aligned} P(I_p + QP) &= (I_n + PQ)P \\ \iff P &= (I_n + PQ)P(I_p + QP)^{-1} \\ \iff (I_n + PQ)^{-1}P &= P(I_p + QP)^{-1} \end{aligned} \quad (\text{A.27})$$

with $I_p + QP$ admitting a right inverse, hence being invertible. $\square$

Proposition A.5 (Generalised Woodbury formula [91]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$. We assume that A , D , and $D + CA^{-1}B$ are invertible. Then $A + BD^{-1}C$ is also invertible, and:*

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \quad (\text{A.28})$$

Proof. We first factorise:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= (A(I + A^{-1}BD^{-1}C))^{-1} \\ &= (I + A^{-1}BD^{-1}C)^{-1}A^{-1} \end{aligned} \quad (\text{A.29})$$

Applying Proposition A.3 to $A^{-1}BD^{-1}C$ yields:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= \left(I - (I + A^{-1}BD^{-1}C)^{-1}A^{-1}BD^{-1}C \right) A^{-1} \\ &= A^{-1} - (I + A^{-1}BD^{-1}C)^{-1}A^{-1}BD^{-1}CA^{-1} \end{aligned} \quad (\text{A.30})$$

Using Proposition A.4 with $P = A^{-1}B$ and $Q = D^{-1}C$ further gives:

$$(I + A^{-1}BD^{-1}C)^{-1}A^{-1}B = A^{-1}B(I + D^{-1}CA^{-1}B)^{-1} \quad (\text{A.31})$$

Therefore:

$$\begin{aligned} (A + BD^{-1}C)^{-1} &= A^{-1} - A^{-1}B(I + D^{-1}CA^{-1}B)^{-1}D^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B(D(I + D^{-1}CA^{-1}B))^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} \end{aligned} \quad (\text{A.32})$$

$\square$

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